Revised course structure and curriculum for B.Sc. Honours in Management and Information Technology Degree Programme (MIT), and B.Sc. Honours in Information Technology (IT), Department of Industrial Management, Faculty of Science

The B.Sc. in Management and Information Technology degree programme and B.Sc. in Information Technology degree programme are categorized as Level 6, 4-year Honours Degree as per the Sri Lankan Qualification Framework.

As per the University Senate approved entry criterion, after the completion of the first year of the programme, students have either the option of reading for the BSc Honours in Management and Information Technology degree programme or the BSc Honours in Information Technology degree programme.

Students reading for the BSc Hons in MIT, will have the option to specialise in three streams. They are

a) Business Systems Engineering (BSE)
b) Operations and Supply Chain Management (OSCM)
c) Information Systems (IS)

The detailed programme structure for the BSc Honours in MIT and BSc Hons in IT are given below.

The reviesed curriculum will be offered from academic year 2022/2023

B.Sc. Honours in Management and Information Technology degree Programme

Table 1. Detailed programme structure – Year of Study 1				
Course Code	**Course**	**Semester**	**Credits**	**Compulsory /Optional**
DELT 11232	English for Professionals	1	2	C
GNCT 11212[a]	Personal Progress Development I	1	2	C
INTE 11213	Fundamentals of Computing	1	3	C
INTE 11223	Programming Concepts	1	3	C
MGTE 11233	Business Statistics and Economics	1	3	C
MGTE 11243	Principles of Management & Organizational Behaviour	1	3	C
PMAT 11212	Discrete Mathematics for Computing I	1	2	C
MGTE 12253	Accounting Concepts and Costing	2	3	C
INTE 12243	Computer Networks	2	3	C
INTE 12213	Object Oriented Programming	2	3	C
INTE 12223	Database Design and Development	2	3	C
MGTE 12263	Optimization Methods in Management Science	2	3	C
MGTE 12273	Industry and Technology	2	3	C
PMAT 12212	Discrete Mathematics for Computing II	2	2	C

Year of Study 2				
Course Code	**Course Title**	**Semester**	**Credits**	**Compulsory /Optional**
INTE 21213	Information Systems Modelling	1	3	C

INTE 21313	Business Information Systems	1	3	C
INTE 21323	Web Applications Development	1	3	C
INTE 21333	Event Driven Programming	1	3	C
MGTE 21243	Marketing Management	1	3	C
MGTE 21233	Operations Management	1	3	C
INTE 21343	Software Engineering Concepts	1	3	C
MGTE 22273	Human Resource Management & Leadership Communication	2	3	C
INTE 22343	Data Structures and Algorithms	2	3	C
INTE 22303	Artificial Intelligence	2	3	C
MGTE 22263	Logistics and Supply Chain Management	2	3	C
INTE 22283	Mobile Applications Development	2	3	C
GNCT 24212[a]	Personal Progress Development II	2	2	C

Year of Study 3				**Compulsory /Optional**		
Course Code	**Course Title**	**Semester**	**Credits**	**BSC**	**OSCM**	**IS**
INTE 31356	Software Development Project	1	6	C	C	C
MGTE 31393	Managerial Finance	1	3	C	C	C
MGTE 31293	Computer Integrated Manufacturing	1	3	C	O	O
MGTE 31403	Management of Technology	1	3	C	O	O
MGTE 31413	Warehouse Management and Industrial Shipping	1	3	O	C	O
MGTE 31373	Project Management	1	3	C	C	C
MGTE 31423	Advanced Optimization Methods in Management Science	1	3	C	C	O
MGTE 31433	Computer based tools for Management Applications	1	3	O	O	O
MGTE 31303	Procurement and Supply Management	1	3	O	C	O
INTE 31423	Data Analytics and Visualization	1	3	O	O	C
INTE 31413	Information Technology Infrastructure	1	3	O	O	C
INTE 31393	Information Security	1	3	O	O	C
MGTE 31383	Research Methods	1	3	O	O	O
MGTE 31443	Strategic Marketing and International Trade	1	3	O	O	O
GNCT 32216	Internship	2	6	C	C	C

Year of Study 4				**Compulsory /Optional**		
Course Code	**Course Title**	**Semester**	**Credits**	**BSC**	**OSCM**	**IS**
INTE 43216[b]	Research Project	1 & 2	6			C
MGTE 43216[b]	Research Project	1 & 2	6	C	C	

MGTE 41323	Professional Practices	1	3	C	C	C
MGTE 41333	Business Process Engineering	1	3	C	O	O
MGTE 41233	Corporate Finance	1	3	O	O	O
INTE 41283	Information Systems Management and Strategy	1	3	O	O	C
MGTE 41303	Enterprise Systems	1	3	O	O	C
MGTE 41313	Statistical Data Modeling	1	3	C	C	C
MGTE 41343	Logistics System Analysis and Geomatics	1	3	O	C	O
MGTE 41353	Sustainability and Economics in Logistics and Supply Chain	1	3	O	O	O
MGTE 41363	Digital Innovations Management	1	3	O	O	O
MGTE 41373	Strategic Management	1	3	O	O	O
MGTE 41383	Advanced operations Management	1	3	O	O	O
MGTE 44273	Innovation & Entrepreneurship	1 OR 2	3	O	O	O
INTE 44303	Information Audit and Assurance	1 OR 2	3	O	O	O
MGTE 42323	Strategic Quality Management and Lean Six Sigma	2	3	O	C	O
MGTE 42213	Industrial and Systems Engineering	2	3	C	O	O
MGTE 42313	Advanced Supply Chain and Logistics Applications	2	3	O	O	O
MGTE 42243	Advanced Planning and Scheduling	2	3	O	O	O
MGTE 42333	Business and Information Technology Law	2	3	O	O	O
MGTE 42223	Investment Management	2	3	O	O	O

[a] Credits will not be considered for GPA calculation
[b] Student will take either of the two modules

Students can consider the course modules mentioned in Table 2and any other course modules offered by the Deparment of Industrial Management, as possible optional modules, and other modules offered by any other department, subject to meeting the prerequisite requirements, subject availability, capacity and scheduling.

1. Award of Degree

Eligibility for the Award of the Bachelor of Science Honours in Management and Information Technology (MIT) Degree

To be eligible for the BSc Hons (MIT) Degree, a student must

(i) accumulate grades of D or better in course units including all compulsory course units, aggregating to at least 72 credits from year of study 1 and year of study 2, and aggregating to at least 132 credits from all four years, of which at least 30 credits must be from each of year of study 3 and year of study 4 academic year separately,
(ii) obtains grades of C or better in course units totalling to at least 105 credits with at least D grades for the remaining course units, and
(iii) obtain grades of C or better for either MGTE 43216 or INTE 43216 course unit and for INTE 31356 and GNCT 32216 course units, and
(iv) Pass GNCT 11212 and GNCT 24212 course units, and
(v) obtain a minimum GPA of 2·00, and
(vi) complete the relevant requirements within a period of six consecutive academic years.

1.1. Award of Classes

a) First Class

A student who is eligible for the BSc Hons (MIT) Degree may be awarded First Class if he/she

(i) obtains grades of C or better in all the course units considered for the calculation of the GPA, and
(ii) obtains grades of A or better aggregating to at least half the number of credits accumulated and considered for the calculation of the GPA, and
(iii) obtains a minimum GPA of 3.70 and
(iv) completes the relevant requirements within four academic years.

Note:

A student who obtains grades of D, D+ and C- for a maximum of 4 credits in compulsory course units, and fulfils all the other requirements stipulated under a) above, may be considered by the Board of Examiners for the award of Second Class (Upper Division).

b) Second Class (Upper Division)

A student who is eligible for the BSc Hons (MIT) Degree may be awarded Second Class (Upper Division) if he/she

(i) obtains grades of C or better in course units, including the compulsory course units, aggregating to at least 117 credits, and grades of D or better in the remaining course units considered for GPA calculation, and
(ii) obtains grades of B or better aggregating to at least half the number of credits accumulated and considered for the calculation of the GPA, and
(iii) obtains a minimum GPA of 3·30 and
(iv) completes the relevant requirements within four academic years

Note:

A student who obtains minimum grades of D, D+ and C- for a maximum of 4 credits in compulsory course units, and fulfils all the other requirements stipulated under b) may be considered by the Board of Examiners for the award of Second Class (Lower Division).

c) Second Class (Lower Division)

A student who is eligible for the BSc Hons (MIT) Degree may be awarded Second Class (Lower Division) provided he/she

(i) obtains grades of C or better in course units, including all compulsory course units, aggregating to at least 117 credits, and grades of D or better in the remaining course units considered for GPA calculation, and
(ii) obtains grades of B or better aggregating to at least half the number of credits accumulated and considered for the calculation of the GPA, and
(iii) obtains minimum GPA of 3·00 and
(iv) completes the relevant requirements within four academic years.

2. Exit Point at the end of year 3 for the Bachelor of Science Honours in Management and Information Technology Degree (MIT)

Eligibility for the award of the Bachelor of Science in Management and Information Technology Degree

To be eligible for the BSc (MIT) Degree, a student must

(i) accumulate grades of D or better in course units including all compulsory course units, aggregating to at least 72 credits from year of study 1 and year of study 2, and aggregating to at least 102 credits from all three years, of which at least 30 credits must be from year of study 3,
(ii) obtains grades of C or better in course units totalling to at least 85 credits with at least D grades for the remaining course units, and
(iii) obtains grades of C or better for INTE 31356 and GNCT 32216 course units, and
(iv) Pass GNCT 11212 and GNCT 24212 course units, and
(v) obtain a minimum GPA of 2·00, and
(vi) complete the relevant requirements within a period of five consecutive academic years.

2.1. Award of Classes

a) First Class

A student who is eligible for the BSc (MIT) Degree may be awarded First Class if he/she

(i) obtains grades of C or better in all the course units considered for the calculation of the GPA, and
(ii) obtains grades of A or better aggregating to at least half the number of credits accumulated and considered for the calculation of the GPA, and
(iii) obtains a minimum GPA of 3·70 and
(iv) completes the relevant requirements within three academic years.

Note:
A student who obtains grades of D, D+ and C- for a maximum of 4 credits in compulsory course units, and fulfils all the other requirements stipulated under a) above, may be considered by the Board of Examiners for the award of Second Class (Upper Division).

b) Second Class (Upper Division)

A student who is eligible for the BSc (MIT) Degree may be awarded Second Class (Upper Division) if he/she

(i) obtains grades of C or better in course units, including the compulsory course units, aggregating to at least 99 credits, and grades of D or better in the remaining course units considered for GPA calculation, and
(ii) obtains grades of B or better aggregating to at least half the number of credits accumulated and considered for the calculation of the GPA, and
(iii) obtains a minimum GPA of 3·30 and
(iv) completes the relevant requirements within three consecutive academic years

Note:
A student who obtains minimum grades of D, D+ and C- for a maximum of 4 credits in compulsory course units, and fulfils all the other requirements stipulated under b) may be considered by the Board of Examiners for the award of Second Class (Lower Division).

c) Second Class (Lower Division)

A student who is eligible for the BSc (MIT) Degree may be awarded Second Class (Lower Division) provided he/she

(i) obtains grades of C or better in course units, including all compulsory course units, aggregating to at least 99 credits, and grades of D or better in the remaining course units considered for GPA calculation, and
(ii) obtains grades of B or better aggregating to at least half the number of credits accumulated and considered for the calculation of the GPA, and
(iii) obtains minimum GPA of 3·00 and
(iv) completes the relevant requirements within three academic years.

3. Fallback option for the Bachelor of Science Honours in Management and Information Technology Degree (MIT)

Eligibility for the award of the Higher Diploma in Management and Information Technology

To be eligible for the Higher Diploma in MIT, a student must

(i) accumulate grades of D or better in course units including all compulsory course units, aggregating to at least 72 credits from year of study 1 and year of study 2, and
(ii) obtains grades of C or better in course units totalling to at least 60 credits with at least D grades for the remaining course units, and
(iii) Pass GNCT 11212 and GNCT 24212 course units, and
(iv) obtain a minimum GPA of 2·00, and
(v) complete the relevant requirements within a period of six consecutive academic years.

B.Sc. Honours in Information Technology Degree Programme

Table 2: Detailed programme structure

Year of Study 1				
Course Code	**Course**	**Semester**	**Credits**	**Compulsory /Optional**
DELT 11232	English for Professionals	1	2	C
GNCT 11212[a]	Personal Progress Development I	1	2	C
INTE 11213	Fundamentals of Computing	1	3	C
INTE 11223	Programming Concepts	1	3	C
MGTE 11233	Business Statistics and Economics	1	3	C
MGTE 11243	Principles of Management & Organizational Behaviour	1	3	C
PMAT 11212	Discrete Mathematics for Computing I	1	2	C
MGTE 12253	Accounting Concepts and Costing	2	3	C
INTE 12243	Computer Networks	2	3	C
INTE 12213	Object Oriented Programming	2	3	C
INTE 12223	Database Design and Development	2	3	C
MGTE 12263	Optimization Methods in Management Science	2	3	C
MGTE 12273	Industry and Technology	2	3	C
PMAT 12212	Discrete Mathematics for Computing II	2	2	C

Year of Study 2				
Course Code	**Course Title**	**Semester**	**Credits**	**Compulsory /Optional**
INTE 21213	Information Systems Modelling	1	3	C
INTE 21313	Business Information Systems	1	3	C
INTE 21323	Web Applications Development	1	3	C
INTE 21333	Event Driven Programming	1	3	C
INTE 21243	Computer Architecture and Operating Systems	1	3	C
INTE 22293	Software Architecture and Process Models	2	3	C
INTE 22343	Data Structures and Algorithms	2	3	C
INTE 22253	Distributed Systems and Cloud Computing	2	3	C
INTE 22263	Embedded Systems Development	2	3	C
INTE 22303	Artificial Intelligence	2	3	C
INTE 22313	Software Design Patterns and Frameworks	2	3	C
INTE 22283	Mobile Applications Development	2	3	C
GNCT 24212[a]	Personal Progress Development II	2	2	C

Year of Study 3				
Course Code	**Course Title**	**Semester**	**Credits**	**Compulsory /Optional**
INTE 31233	Human Computer Interaction	1	3	C
INTE 31393	Information Security	1	3	C
INTE 31356	Software Development Project	1	6	C
MGTE 31373	Project Management	1	3	C
INTE 31243	Software Quality Engineering	1	3	O
INTE 31283	Big Data and Data Warehousing	1	3	O
INTE 31373	Machine Learning	1	3	O
INTE 31403	System Administration and Maintenance	1	3	O
MGTE 31383	Research Methods	1	3	O
PMAT 31212	Mathematics for Computing - 3	1	2	O
GNCT 32216	Internship	2	6	C

Year of Study 4				
Course Code	**Course Title**	**Semester**	**Credits**	**Compulsory /Optional**
INTE 43216[b]	Research Project	1 & 2	6	C
MGTE 41313	Statistical Data Modeling	1	3	C
MGTE 41323	Professional Practices	1	3	C
INTE 41393	System Integration Technologies	1	3	C
INTE 41373	Image Processing	1	3	O
INTE 41323	Neural Networks and Deep Learning	1	3	O
INTE 41383	Industrial Automation	1	3	O
INTE 41403	Advanced Databases	1	3	O
INTE 41413	Internet of Things	1	3	O
MGTE 42303	Business and Information Technology Law	2	3	C
INTE 42353	Semantic Web and Ontological Engineering	2	3	O
INTE 42333	Complex Systems and Agent Based Modelling	2	3	O
INTE 42343	Natural Language Processing	2	3	O

[a] Credits will not be considered for GPA calculation
[b] Student will take either of the two modules

Students can consider the course modules mentioned in Table 1 and any other course modules offered by the Deparment of Industrial Management, as possible optional modules, and other modules offered by any other department, subject to meeting the prerequisite requirements, subject availability, capacity and scheduling.

4. Award of Degree

Eligibility for the Award of the Bachelor of Science Honours in Information Technology (IT) Degree

To be eligible for the BSc Hons in IT Degree, a student must

(i) accumulate grades of D or better in course units including all compulsory course units, aggregating to at least 72 credits from year of study 1 and yeaer of study 2, and aggregating to at least 132 credits from all four years, of which at least 30 credits must be from each of year of study 3 and year of study 4 academic year separately,
(ii) obtains grades of C or better in course units totalling to at least 105 credits with at least D grades for the remaining course units, and
(iii) obtain grades of C or better for INTE 43216 course unit and for INTE 31356 and GNCT 32216 course units, and
(iv) Pass GNCT 11212 and GNCT 24212 course units, and
(v) obtain a minimum GPA of 2·00, and
(vi) complete the relevant requirements within a period of six consecutive academic years.

4.1. Award of Classes

a) First Class

A student who is eligible for the BSc Hons in IT Degree may be awarded First Class if he/she

(i) obtains grades of C or better in all the course units considered for the calculation of the GPA, and
(ii) obtains grades of A or better aggregating to at least half the number of credits accumulated and considered for the calculation of the GPA, and
(iii) obtains a minimum GPA of 3.70 and
(iv) completes the relevant requirements within four academic years.

Note:
A student who obtains grades of D, D+ and C- for a maximum of 4 credits in compulsory course units, and fulfils all the other requirements stipulated under a) above, may be considered by the Board of Examiners for the award of Second Class (Upper Division).

b) Second Class (Upper Division)

A student who is eligible for the BSc Hons in IT Degree may be awarded Second Class (Upper Division) if he/she

(i) obtains grades of C or better in course units, including the compulsory course units, aggregating to at least 117 credits, and grades of D or better in the remaining course units considered for GPA calculation, and
(ii) obtains grades of B or better aggregating to at least half the number of credits accumulated and considered for the calculation of the GPA, and
(iii) obtains a minimum GPA of 3·30 and
(iv) completes the relevant requirements within four academic years

Note:
A student who obtains minimum grades of D, D+ and C- for a maximum of 4 credits in compulsory course units, and fulfils all the other requirements stipulated under b) may be considered by the Board of Examiners for the award of Second Class (Lower Division).

c) Second Class (Lower Division)

A student who is eligible for the BSc Hons in IT Degree may be awarded Second Class (Lower Division) provided he/she

(i) obtains grades of C or better in course units, including all compulsory course units, aggregating to at least 117 credits, and grades of D or better in the remaining course units considered for GPA calculation, and
(ii) obtains grades of B or better aggregating to at least half the number of credits accumulated and considered for the calculation of the GPA, and
(iii) obtains minimum GPA of 3·00 and
(iv) completes the relevant requirements within four academic years.

5. Exit Point at the end of Year 3 for the Bachelor of Science Honours in Information Technology Degree (IT)

Eligibility for the award of the Bachelor of Science in Information Technology Degree

To be eligible for the BSc in IT Degree, a student must

(i) accumulate grades of D or better in course units including all compulsory course units, aggregating to at least 72 credits from year of study 1 and year of study 2, and aggregating to at least 102 credits from all three years, of which at least 30 credits must be from year 3,

(ii) obtains grades of C or better in course units totalling to at least 85 credits with at least D grades for the remaining course units, and

(iii) obtains grades of C or better for INTE 31356 and GNCT 32216 course units, and

(iv) Pass GNCT 11212 and GNCT 24212 course units, and

(v) obtain a minimum GPA of 2·00, and

(vi) complete the relevant requirements within a period of five consecutive academic years.

5.1. Award of Classes

a) First Class

A student who is eligible for the BSc in IT Degree may be awarded First Class if he/she

(i) obtains grades of C or better in all the course units considered for the calculation of the GPA, and

(ii) obtains grades of A or better aggregating to at least half the number of credits accumulated and considered for the calculation of the GPA, and

(iii) obtains a minimum GPA of 3·70 and

(iv) completes the relevant requirements within three academic years.

Note:

A student who obtains grades of D, D+ and C- for a maximum of 4 credits in compulsory course units, and fulfils all the other requirements stipulated under a) above, may be considered by the Board of Examiners for the award of Second Class (Upper Division).

b) Second Class (Upper Division)

A student who is eligible for the BSc in IT Degree may be awarded Second Class (Upper Division) if he/she

(i) obtains grades of C or better in course units, including the compulsory course units, aggregating to at least 99 credits, and grades of D or better in the remaining course units considered for GPA calculation, and

(ii) obtains grades of B or better aggregating to at least half the number of credits accumulated and considered for the calculation of the GPA, and

(iii) obtains a minimum GPA of 3·30 and

(iv) completes the relevant requirements within three consecutive academic years

Note:

A student who obtains minimum grades of D, D+ and C- for a maximum of 4 credits in compulsory course units, and fulfils all the other requirements stipulated under b) may be considered by the Board of Examiners for the award of Second Class (Lower Division).

c) Second Class (Lower Division)

A student who is eligible for the BSc in IT Degree may be awarded Second Class (Lower Division) provided he/she

(i) obtains grades of C or better in course units, including all compulsory course units, aggregating to at least 99 credits, and grades of D or better in the remaining course units considered for GPA calculation, and

(ii) obtains grades of B or better aggregating to at least half the number of credits accumulated and considered for the calculation of the GPA, and

(iii) obtains minimum GPA of 3·00 and

(iv) completes the relevant requirements within three academic years.

6. Fallback option for the Bachelor of Science Honours in Information Technology Degree (IT)

Eligibility for the award of the Higher Diploma in Information Technology

To be eligible for the Higher Diploma in IT, a student must

(i) accumulate grades of D or better in course units including all compulsory course units, aggregating to at least 72 credits from year of study 1 and year of study 2, and
(ii) obtains grades of C or better in course units totalling to at least 60 credits with at least D grades for the remaining course units, and
(iii) Pass GNCT 11212 and GNCT 24212 course units, and
(iv) obtain a minimum GPA of 2·00, and
(v) complete the relevant requirements within a period of six consecutive academic years.

Detailed Course Modules:

a) Information Technology

Year of Study 1:

Year of Study	1		
Semester	1		
Course Code	INTE 11213		
Course Name	Fundamentals of Computing		
Pre-requisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical/ Tutorial	Self-Guided Study
	45	30	75

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Describe the evolution of the computer
2. Describe the main components of a computer system
3. Distinguish between application and system software
4. Explain how data and program are represented in computers
5. Explain how combinational circuits make logical decisions in computers
6. Explain how sequential circuits perform sequence of actions in computers
7. Define basic computer architectures and operations

Course Content:
Evolution of computers, main components of a computer system, application software, system software, computer hardware, connectivity and communication systems, data and program representation, digital logic, combinational circuits, sequential circuits, computer architectures

Teaching /Learning Methods:
Lectures, supervised hands on practical/tutorial sessions, guided self-study assignments and group work

Assessment Criteria:
End semester examination and continuous assessments

Continuous Assessment 0 - 25%	Final Assessment 75 - 100%		
Details: quizzes, mid-term, assignments 0 - 25%, 0 - 25%, 0 - 25%	Theory (%) 100%	Practical (%) 0%	Other (%) (specify)

Recommended Reading:

1. Timothy O'Leary, Linda O'Leary and Daniel O'Leary, (2022), Computing Essentials 2023, 29th edition, McGraw–Hill
2. William Stallings, (2018), Computer Organization and Architecture: Designing for Performance, 11th Edition, Pearson
3. William Stallings, (2017), Operating Systems: Internals and Design Principles, 9th Edition, Pearson
4. Material provided in CAL.

Year of Study	1		
Semester	1		
Course Code	INTE 11223		
Course Name	Programming Concepts		
Pre-requisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical/ Tutorial	Self-Guided Study
	30	45	75

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Explain the fundamental software components of a computer system
2. Design an algorithm to solve a problem
3. Develop programs using syntax and semantics of a general-purpose programming language
4. Apply the principles of program design to implement practical application

Course Content:
Problem solving process, program execution cycle, language translation, elementary data types, variables, constants, type casting, control structures, functions, recursion, structured data types, abstract data types, references, pointers, static and dynamic memory allocations, file handling, software version control, programming standards and best practices.

Teaching /Learning Methods:
Lectures, supervised hands on practical sessions and group work

Assessment Criteria:
End-of-semester examination and continuous assessment

Continuous Assessment 40 - 60%	Final Assessment 40 - 60%		
Details: quizzes %, mid-term %, Assignments% 0 - 20 %, 0 – 20 % 0 – 40 %	Theory (%) 50%	Practical (%) 50%	Other (%)(specify)

Recommended Reading:

1. Paul J. Deitel and Harvey M. Deitel, (2017), C++ How to Program, 10th Edition, Pearson.
2. Bjarne Stroustrup, (2014), Programming: Principles and Practice Using C++, 2nd Edition, Addison-Wesley Professional.
3. Siddhartha Rao, (2017), Teach Yourself C++ in One Hour a Day, 8th Edition, Sams Publishing.
4. Materials provided in the CAL.

Year of Study	1		
Semester	2		
Course Code	INTE 12243		
Course Name	Computer Networks		
Pre-requisites	INTE 11213 - Fundamentals of Computing		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	30	45	75

Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Explain layers of abstraction in computer network protocol stack, and how these facilitate data communication between hosts in a network.
2. Use industry-level networking tools to simulate different network configurations, analyse network data, networking protocols and their operations, and provide networking solutions satisfying real-world requirements.
3. Reflect on how network hardware and protocols are implemented and apply that knowledge in analysing real-world network issues related to network performance, scalability, and security.

Course Content:
Overview of data communication, network topologies and types, network devices and components, OSI and TCP/IP models, network protocols, physical layer functions, transmission media, signal encoding and modulation, data link layer functions, framing and link layer addressing, Ethernet and LAN technologies, Virtual LANs, network layer functions, IP addressing and subnetting, network address translation, static and dynamic routing, transport layer functions, TCP and UDP, flow control and error handling, application layer functions, application layer protocols and sockets, DNS and DHCP, wireless networks, network performance, network monitoring and SNMP, security threats and attacks, cryptography and encryption, firewalls, intrusion detection systems, virtual private networks, software defined networks, the Internet and Internet service providers.

Teaching /Learning Methods:
Lectures, supervised practical sessions, and group work.

Assessment Criteria:
End-of-semester examination and continuous assessment.

Continuous Assessment: 40 - 60%	Semester-End Examination: 40 - 60%		
Details: Mid-term and end-term quizzes: 20 – 30% Assessments: 20 – 30%	Theory (%) 40 – 50%	Practical (%) 20 – 30%	Other (%) N/A

Recommended Reading:

1. F James, F Kurose, Amherst Keith Ross, 2017, Computer Networking: A Top-Down Approach, 7th edition, Pearson.
2. W Odom, 2019, CCNA 200-301 Official Cert Guide Volume 1, Cisco Press.
3. W Stalling, 2014, Data and Computer Communications, 10th Edition, Pearson Education.
4. Materials provided in the CAL.

Year of Study	1		
Semester	2		
Course Code	INTE 12213		
Course Name	Object Oriented Programming		
Pre-requisites	INTE 11223 - Programming Concepts		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical/ Tutorial	Self-Guided Study
	30	45	75

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Describe the principles of object-oriented programming with examples
2. Discuss the advantages of object-oriented approach
3. Design, develop, test and debug object oriented computer programs and monitor performance
4. Develop GUI using an object oriented programming language
5. Develop object oriented software applications with database connectivity

Course Content:
Principles of Object Oriented Programming (OOP), Classes and objects and its methods, OOP principles implementation in Java (information hiding and encapsulation, inheritance, interfaces, abstraction, polymorphism, static and dynamic binding), Interfaces and Abstract classes, Exception Handling, Debugging and Testing, Garbage Collection, GUI Components, Generic Classes and Methods, Accessing Databases with JDBC, Multithreading, Version controlling tools setup (GitHub/GitLab/Bitbucket etc)

Teaching /Learning Methods:
Lectures, supervised hands on practical sessions and group work

Assessment Criteria:
End-of-semester examination and continuous assessment

Continuous Assessment	Final Assessment		
40 - 60%	40 - 60%		
Details: quizzes %, mid-term %, Assignments% 0 - 20 %, 0 – 20 % 0 – 40 %	Theory (%) 50%	Practical (%) 50%	Other (%)(specify)

Recommended Reading:

1. Matt Weisfeld, (2019), Object-Oriented Thought Process,5th Edition , Addison-Wesley Professional.
2. Deitel, P J and Deitel, H M, (2017), Java How to Program: Early Objects, Pearson, Pearson; 11th edition.
3. Savitch W., (2017), Java: An Introduction to Problem Solving and Programming, Pearson
4. Iuliana Cosmina, (2018), Java for Absolute Beginners: Learn to Program the Fundamentals the Java 9+ Way, 1st edition, Apress.
5. Materials provided in the CAL.

Year of Study	1		
Semester	2		
Course Code	INTE 12223		
Course Name	Database Design and Development		
Pre-requisites	INTE 11223 - Programming Concepts		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical/ Tutorial	Self-Guided Study
	30	45	75

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Identify data / information needs of an organization
2. Describe the techniques used in storing and retrieving data
3. Design logical data models
4. Develop databases using relational model
5. Define and manipulate data using Structured Query Language (SQL)
6. Recognize new trends in databases

Course Content:
Data and information, database approach, role of database management systems in organizations, different database architectures, relational database design and development: ER modeling, normalization, Structured Query Language (SQL): DDL, DML, new trends in databases

Teaching /Learning Methods:
Lectures, supervised hands on practical sessions, tutorials and group work

Assessment Criteria:
End-of-semester examination and continuous assessment

Continuous Assessment 20-40%	Final Assessment 60 – 80%		
Details: quizzes %, mid-term %, Assignments% 0 – 10 %, 0 – 10 % 0 – 20 %	Theory (%) 100%	Practical (%)	Other (%)(specify)

Recommended Reading:

1. Ramez Elmasri and Shamkant B. Navathe, (2017), Fundamentals of Database Systems, 7th Edition, Pearson.
2. Jeffrey A. Hoffer, Ramesh Venkataraman and Heikki Topi, (2019), Modern Database Management, 13th Edition, Pearson.
3. C. Coronel and S. Morris, (2018), Database Systems: Design, Implementation and Management, 13th Edition, Cengage Learning
4. G. Harrison, (2015), Next Generation Databases: NoSQL and Big Data, Apress
5. Materials provided in the CAL.

Year of study 2:

Year of Study	2		
Semester	1		
Course Code	INTE 21213		
Course Name	Information Systems Modelling		
Pre-requisites	INTE 11223- Programming Concepts INTE 12213- Object Oriented Programming INTE 12223- Database Design and Development		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown	Theory	Practical	Guided Self-study
(Notional Hrs.)	45	-	105

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Understand human and technical factors involved in information systems analysis and design
2. Understand the importance of system modelling and modelling process
3. Apply the object oriented and structured modelling approaches.
4. Apply software modelling tools effectively in developing information systems
5. Construct real world Information System development models based on object oriented and structured approaches.

Course Content:
System Concepts: system elements and its environment, Software Development Life Cycle (SDLC), Evolution of system analysis & design, facts gathering techniques, Software modelling: structured approach and object-oriented approach, Unified Modelling language (UML): Overview of the UML notation, UML diagrams, requirements modelling, Modelling real world problems and solutions

Teaching /Learning Methods:
Lectures, assignments and group project

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment	Final Assessment		
20 – 40%	60 – 80%		
Details: quizzes %, mid-term %, Assignments% 0 – 10 %, 0 – 10 % 0 – 20 %	Theory (%) 100%	Practical (%) 0%	Other (%)(specify)

Recommended Reading:

1. Martina Seidl, Marion Scholz, Christian Huemer and Gerti Kappel, (2015), UML @ Classroom: An Introduction to Object-Oriented Modeling, Springer Publishing, ISBN 3319127411
2. Gomaa, H, (2011), Software Modeling and Design: UML, Use Cases, Patterns, and Software Architectures. Cambridge University Press.
3. Valacich, J, George, J, Hoffer, J, (2015), Essentials of Systems Analysis and Design, 6th Edition, Pearson
4. Kendall & Kendall, (2023), System Analysis and Design, 11th Edition, Pearson
5. Material provided in CAL.

Year of Study	2		
Semester	1		
Course Code	INTE 21313		
Course Name	Business Information Systems		
Pre-requisites	INTE 11213 - Fundamentals of Computing		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	30	45	75

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Understand the fundamental concepts and the role of information in the enterprise
2. Explain different systems that support enterprises to operate and compete effectively
3. Explain major trends in IT-based infrastructures and assess their impact on an enterprise
4. Assess how information systems support enterprises to achieve a strategic advantage
5. Examine the information security challenges faced by enterprises and Evaluate the tools and technologies for safeguarding information resources
6. Examine ethical and social challenges posed by information technology

Course Content:
Role of IT in organizations; Information systems in enterprises; Strategic advantage with information technology; IT infrastructure of an enterprise; Information security challenges; Policies, tools and technologies for safeguarding information assets; Ethical and social challenges associated with information technology

Teaching /Learning Methods:
Lectures, supervised hands-on practical sessions, and group work

Assessment Criteria:
End-of-semester examination and continuous assessment

Continuous Assessment 40 – 60%	Semester-End Examination 60 – 40%		
Details: Assignments: 50% MCQs: 50%	Theory (%) 100%	Practical (%)	Other (%)(specify)

Recommended Reading:

1. Kenneth C Laudon and Jane P Laudon, 2022, Management Information Systems, 17th edition, Pearson
2. Baltzan, P., 2021, Business driven information systems, 7th edition, McGraw Hill
3. David Sutton et al., (2021), Information Security Management Principles, 3rd edition, BCS
4. Michael E. Whitman and Herbert J. Mattord, 2021, Principles of Information Security, 7th Edition, Cengage Learning
5. Materials provided in the LMS.

Year of Study	2		
Semester	1		
Course Code	INTE 21323		
Course Name	Web Applications Development		
Pre-requisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	30	45	75

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Explain the relevance, purpose, and operation of technologies of web applications.
2. Design and develop client-side and server-side components of a web-based application using contemporary tools, languages, and frameworks.
3. Design and develop a dynamic, secure, responsive web-based applications.
4. Identify options for deploying web applications in a suitable production environment.

Course Content:
Web application architecture, client-server model, HTML, CSS, client-side scripting, Document Object Model, server-side development, working with databases, error handling and validation, managing state, asynchronous web applications, AJAX, WebSockets, advanced CSS and responsive design, Model-View Controller (MVC) architecture, Web Application Hosting (traditional hosting solutions, cloud hosting, etc), Containerization on web application.

XML Processing and Web Services, API development, documentation and testing, Java Web Software: Servlets and JSP, Web Application Security, Web Monitoring and Analytics, Search Engine Optimization and Social Network Integration.

Teaching /Learning Methods:
Lectures, supervised hands-on practical sessions, and group work

Assessment Criteria:
End-of-semester examination and continuous assessment

Continuous Assessment	Semester-End Examination		
40 – 60%	40 – 60%		
Details: Two problem solving reports: 20% each Structured Examination Paper: 60%	Theory (%) 50%	Practical (%) 50%	Other (%)(specify)

Recommended Reading:

1. R. Connolly and R. Hoar, Fundamentals of Web Development, 3rd Edition, Pearson, 2021
2. P. Deitel and H. Deitel, A. Deitel, Internet & World Wide Web How to Program, 5th Edition, Pearson, 2021
3. L. Welling and L. Thomson, PHP & MySQL Web Development. Addison-Wesley, 2017
4. Material provided in CAL.

Year of Study	2		
Semester	1		
Course Code	INTE 21333		
Course Name	Event Driven Programming		
Pre-requisites	INTE 12223 – Database Design and Development, INTE 12213 – Object Oriented Programming		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown	Theory	Practical	Guided Self-study
(Notional Hrs.)	30	45	75

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Explain the difference between event-driven and command-line programming.
2. Explain the use of object-oriented programming concepts in event-driven programming.
3. Choose appropriate components and develop quality and friendly user interfaces.
4. Choose appropriate events and develop highly interactive applications.
5. Design and develop interactive reports to visualise data.
6. Develop interactive systems to solve complex, real-world problems.

Course Content:
Introduction to event-driven programming, event-driven model, event detection and handling, Graphical User Interface (GUI) design, handling multiple events, design of reports and data visualization, building and publishing interactive multiuser applications, event driven programming design patterns and frameworks. emerging trends.

Teaching /Learning Methods:
Lectures, supervised practical sessions and guided self-study assignments

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment 50%	Final Assessment 60%		
Details: Individual Project 40 %	Theory (%) 100%	Practical (%) 0%	Other (%)(specify)

Recommended Reading:

1. Paul Deitel and Harvey Deitel, (2016), Visual C# How to Program, 6th Edition, Pearson.
2. John Sharp, (2022), Microsoft Visual C# Step by Step, 10th Edition, Microsoft Press.
3. Material provided in CAL.

Year of Study	2		
Semester	1		
Course Code	INTE 21343		
Course Name	Software Engineering Concepts		
Pre-requisites	INTE 21213 - Information Systems Modelling		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown	Theory	Practical	Guided Self-study
(Notional Hrs.)	45	30	75

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Explain the stages and activities carried out in the software development life cycle
2. Discuss the different system development models and methodologies
3. Explain the different tools and techniques for systems development
4. Select appropriate methods and techniques to produce a software to a given scenario
5. Identify the requirements and prepare system requirement specification
6. Use appropriate modelling techniques to produce a system design for a given scenario
7. Explain development and testing approaches to implement the software for the given scenario

Course Content:
Software engineering processes, agile software development, requirement engineering, system modelling, architectural design, design and implementation, software testing, software evolution, dependable systems, reliability engineering, security engineering, software reuse, systems engineering, systems of systems, quality management, configuration management.

Teaching /Learning Methods:
Lectures, supervised hands on practical sessions and group work

Assessment Criteria:
End-of-semester examination and continuous assessment

Continuous Assessment	Final Assessment		
20- 40%	60 - 80%		
Details: quizzes %, mid-term %, Assignments% 0 - 20 %, 0 – 20 % 0 – 40 %	Theory (%) 100%	Practical (%) 0%	Other (%)(specify)

Recommended Reading:

1. Pressman R (2019), *Software Engineering - A Practitioner's Approach,* 9th Edition, McGraw Hill.
2. Sommerville I (2018), *Software Engineering*, 10th Edition, Pearson India
3. Material provided in CAL

Year of Study	2		
Semester	1		
Course Code	INTE 21243		
Course Name	Computer Architecture and Operating Systems		
Pre-requisites	INTE 11213 - Fundamentals of Computing; INTE 11223 - Programming Concepts		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown	Theory	Practical/ Tutorial	Self-Guided Study
(Notional Hrs.)	75	0	75

Course Aim/Intended Learning Outcomes:

After completing this module, the students should be able to,

1. Compare and contrast different computer architectures
2. Describe instruction set architecture
3. Describe the functionality and working of the building blocks of computers
4. Demonstrate an understanding of programming in assembly languages
5. Explain the key roles played by an operating system
6. Describe the concepts, models and approaches involved in design of operating systems
7. Utilize knowledge of computer architecture and operating systems in computer programming

Course Content:

Computer architectures, processor architectures, instruction set architecture, stored program approach, fetch-execute cycle, instruction pipelining, operand addressing, microcode, parallelism, machine language, assembly languages, memory and storage, bus architectures, I/O architectures, architectural design constraints, operating systems, need of an operating system, roles of an operating system, multi-programming, time sharing, process management, memory management, file management, notable computer architectures and operating systems, current trends and future insights

Teaching /Learning Methods:

Lectures, supervised hands on practical sessions on assembly programming, and guided self-study assignments

Assessment Criteria:

End of course unit examination and continuous assessments

Continuous Assessment	Final Assessment		
0 – 25%	75 – 100%		
Details: quizzes, mid-term, assignments 0 – 25 %, 0 – 25 %, 0 – 25 %	Theory (%) 100%	Practical (%) 0%	Other (%) (specify)

Recommended Reading:

1. Douglas E. Comer (2017), Essentials of Computer Architecture, 2nd edition, Chapman and Hall/CRC
2. William Stallings (2018), Computer Organization and Architecture: Designing for Performance, 11th Edition, Pearson
3. William Stallings (2017), Operating Systems: Internals and Design Principles, 9th Edition, Pearson
4. Materials provided in the CAL.

Year of Study	2		
Semester	2		
Course Code	INTE 22293		
Course Name	Software Architecture and Process Models		
Pre-requisites	INTE 21213 - Information Systems Modelling		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical/ Tutorial	Self-Guided Study
	45	30	75

Course Aim/Intended Learning Outcomes:

On completion of this course, the student should be able to:

1. Describe software development process models
2. Select suitable software development process model, and tools for software development
3. Apply system thinking and design thinking
4. Describe the importance of software architecture
5. Identify essential elements in software architecture
6. Describe different architecture Patterns
7. Describe quality attributes and basic rules of software design
8. Evaluate the effect of design on software quality
9. Design a software architecture to satisfy the system requirement

Course Content:

Software process models, system thinking and design thinking: generation of new ideas, problem solving and decision making, software architecture and its importance, role of the architect, architectural drivers, software architectural styles, design suitable software architecture for a given system, architecture vs design, the factors that affect designs, the effect of design on software quality, quality attributes of software design, basic rules of software design, design process: the input and start point of designs, the outcome and results of designs, transformation of data, satisfying and discovering constraints

Teaching /Learning Methods:

Lectures, supervised hands on practical sessions, tutorials and group work

Assessment Criteria:

End-of-semester examination and continuous assessment

Continuous Assessment	Final Assessment		
40 – 60%	40 – 60%		
Details: quizzes %, mid-term %, Assignments% 0 – 20 %, 0 – 20 % 0 – 40 %	Theory (%) 100%	Practical (%) 0%	Other (%)(specify)

Recommended Reading:

1. Humberto Cervantes, Rick Kazman (2016), Designing Software Architectures: A Practical Approach (SEI Series in Software Engineering) 1st Edition, Addison-Wesley Professional
2. Mark Richards, Neal Ford (2020),Fundamentals of Software Architecture: An Engineering Approach 1st Edition by O'Reilly Media
3. Zhu, H. (2005), Software Design Methodology: From Principles to Architectural Styles, Butterworth-Heinemann.
4. Braude, E.J. (2003), Software Design: From Programming to Architecture, John Wiley.
5. Sommerville, I. (2017), Software Engineering, 10th Edition, Pearson.
6. Materials provided in the CAL.

Year of Study	2		
Semester	2		
Course Code	INTE 22343		
Course Name	Data Structures and Algorithms		
Prerequisites	INTE 11223 - Programming Concepts INTE 12213 - Object Oriented Programming		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	45	30	75

Course Aim/Intended Learning Outcomes:

After completing this module, the students should be able to,

1. Identify the properties of a good algorithm and a good program
2. Determine and analyse the efficiency of often used algorithms
3. Design correct and efficient algorithms for solving real-world problems
4. Implement data structures often used in computer programming
5. Discuss the real-world applications of different data structures
6. Apply data structures for solving real-world problem
7. Design and develop efficient computer programs using data structures and algorithms

Course Content:

Algorithms and programs, correctness and efficiency of an algorithm, space complexity, time complexity, the best, average and worst cases of time complexity, linear, logarithmic, linear logarithmic, quadratic, polynomial, exponential and factorial functions of time complexity, asymptotic notations, classical searching and sorting algorithms, divide-and-conquer approach, recursion, atomic and composite data types, fundamental data structures, advanced data structures, real-world applications.

Teaching /Learning Methods:

Lectures, supervised hands on practical sessions, and guided self-study assignments

Assessment Criteria:

End semester examination and continuous assessments

Continuous Assessment	Final Assessment		
0 – 25%	75 – 100%		
Details: quizzes, mid-term, assignments 0 – 25 %, 0 – 25 %, 0 – 25 %	Theory (%) 100%	Practical (%) 0%	Other (%) (specify)

Recommended Reading:

1. Sartaj Sahni, (2004), Data Structures, Algorithms and Applications in C++, Silicon Press
2. Richard F. Gilberg and Behrouz A. Forouzan, (2005), Data Structures: A Pseudocode Approach with C++, 2nd edition, Course Technology
3. Steven S. Skiena, (2020), The Algorithm Design Manual, 3rd Edition, Springer-Verlag
4. Material provided in CAL.

Year of Study	2		
Semester	2		
Course Code	INTE 22253		
Course Name	Distributed Systems and Cloud Computing		
Pre-requisites	INTE 12213 – Objected Oriented Programming, INTE 21323 – Web Applications Development		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	30	45	75

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Explain important characteristics of distributed systems.
2. Describe architectural and fundamental models of distributed systems.
3. Explain and compare strategies for interposes communication.
4. Explain the core concepts of the cloud computing paradigm.
5. Apply fundamental concepts in cloud infrastructures to deploy cloud applications that are resilient, flexible, and cost-efficient.
6. Analyze various cloud programming models and apply them to solve problems on the cloud.

Course Content:
Fundamental models for distributed systems, inter process communication and how to handle synchronization, real/logical/virtual time, consistency, replication, fault tolerance and security of distributed file systems, fundamental of cloud computing, cloud infrastructures, virtualization, cloud management techniques and cloud software deployment, cloud storage technologies and relevant distributed file systems, Distributed Transaction, Multi Agent, Distributed Search Engines, P2P systems, Block chain technologies

Teaching /Learning Methods:
Lectures, group work

Assessment Criteria:
End-of-course exam, Assignment, quizzes, mini-project,

Continuous Assessment 40 – 60%	Final Assessment 40 – 60%		
Details: Assignments% , mini-project%, quizzes% 0 – 20 %, 0 – 20 %, 0 – 20 %	Theory (%) 50%	Practical (%) 50%	Other (%)(specify)

Recommended Reading:

1. George Coulouris, Jean Dollimore, Tim Kindberg and Gordon Blair, (2011) Distributed Systems: Concepts and Design, 5th Edition, Pearson.
2. Andrew S. Tanenbaum and Maarten Van Steen, (2007), Distributed Systems: Principles and Paradigms, 2nd Edition, Pearson.
3. Ray J. Rafaels, (2015), Cloud Computing: From Beginning to End, CreateSpace Independent Publishing Platform.
4. Thomas Erl, Zaigham Mahmood, Ricardo Puttini, (2013) Cloud Computing: Concepts, Technology & Architecture, Pearson.
5. Ahmed Banafa (2022) Blockchain Technology and Applications, River Publishers
6. Materials provided in the CAL.

Year of Study	2		
Semester	2		
Course Code	INTE 22263		
Course Name	Embedded Systems Development		
Pre-requisites	INTE 11223 - Programming Concepts INTE 21243 - Computer Architecture and Operating Systems		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown	Theory	Practical/ Tutorial	Self-Guided Study
(Notional Hrs.)	30	45	75

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Define embedded system
2. Discuss the performance requirements and constraints of embedded systems
3. Describe the architecture of embedded systems
4. Describe the core components of an embedded system
5. Compare and contrast single-chip vs single-board computers-based implementations
6. Explain the hardware and software co-design of embedded systems
7. Evaluate different platforms available for embedded systems development
8. Design, develop and debug embedded systems to automate real world tasks

Course Content:
Embedded system, physical computing, requirements and constraints, evolution, embedded system architectures, single-chip and single-board computers, microcontrollers, microprocessors, special function registers, general purpose and input/output registers, transducers, hardware software co-design, firmware development, hardware programmers, bootloaders, embedded operating systems, Analog to Digital Conversion, polling and interrupt handling, timers and counters, Pulse Width Modulation, asynchronous and synchronous data communication, data acquisition and logging, hardware development, resource-aware firmware development, hardware programming, debugging and simulation, power management, performance evaluation, optimization, current trends and future insights

Teaching /Learning Methods:
Lectures, supervised hands on practical sessions, and guided self-study assignments

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment 0 – 25%	Final Assessment 75 – 100%		
Details: quizzes %, mid-term %, Assignments 0 – 25 %, 0 – 25 %, 0 – 25 %	Theory (%) 50%	Practical (%) 50%	Other (%) (specify)

Recommended Reading:

1. Muhammad Ali Mazidi, Sarmad Naimi and Sepehr Naimi (2017), AVR Microcontroller and Embedded Systems: Using Assembly and C, 2nd edition, MicroDigitalEd.
2. Elliot Williams (2014), Make: AVR Programming, Maker Media.
3. Timothy S. Margush (2011), Some Assembly Required: Assembly Language Programming with the AVR Microcontroller, CRC Press.
4. Materials provided in the CAL.

Year of Study	2		
Semester	2		
Course Code:	INTE 22303		
Course Name:	Artificial Intelligence		
Credit Value:	3		
Pre-Requisites	INTE 11223 - Programming Concepts		
Co-Requisites	None		
Hourly Breakdown	Theory	Practical	Independent Learning
	125	0	25

Course Aim/Intended Learning Outcomes:
At the completion of this course student will be able to:

1. Explain the promises and achievements of artificial intelligence during its evolution
2. Appreciate artificial intelligence as a key technology in software engineering
3. Explain the basic artificial intelligence techniques
4. Discuss various applications of artificial intelligence in business and society
5. Implement intelligent agents to solve practical problems
6. Describe the importance of responsible use of artificial intelligence

Course Content:
Introduction to artificial intelligence; Intelligent agents; Solving problems by search; Evolutionary computing and genetic algorithms; Applications of Genetic Algorithms; Knowledge representation and expert systems; Fuzzy logic systems; Introduction to machine learning; Introduction to data analytics; Machine learning applications in data analytics; Foundations of artificial neural networks; Common applications of neural networks; Societal impact of artificial intelligence

Teaching /Learning Methods:

Lectures, Business case discussions, Group assignments and Practical sessions

Assessment Strategy:

Continuous Assessment 25 - 40 %	Final Assessment 60 - 75 %		
Details: Assignments 100 %	Theory (%) 100	Practical (%) 0%	Other (%) (specify) N/A

References/Reading Materials:

1. Russel, S. J., Norwig, P., Artificial Intelligence: A modern Approach, 4E, Pearson, 2020
2. Joshi, P., Artasanchez, A., Artificial Intelligence with Python: Your complete guide to building intelligent apps using Python 3.x, 2E, Packt Publishing, 2020
3. Materials provided in CAL

Year of Study	2		
Semester	2		
Course Code	INTE 22283		
Course Name	Mobile Applications Development		
Pre-requisites	INTE 12213 – Object Oriented Programming INTE 21323 – Web Applications Development		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	30	45	75

Course Aim/Intended Learning Outcomes:

After completing this module, the students should be able to,

1. Describe the characteristics of mobile networks & communication.
2. Choose a suitable mobile development platform based on requirements.
3. Design a proper flow of user interfaces for a mobile application.
4. Develop and test mobile applications.
5. Enforce proper security mechanisms to mobile apps.
6. Work with mobile iot devices.

Course Content:

Cellular networks and mobile networking, mobility of computers, mobile operating systems, mobile application development platforms, mobile architecture, Integrated Development Environments for mobile application development, introduction to mobile UI design patterns, SOAP and REST services integration, introduction to location services, mobile application monetization, mobile application testing, use of smartphone sensors, database Connectivity, hybrid mobile apps, mobile app frameworks, mobile security, IoT with mobile computing

Teaching /Learning Methods:

Lectures with practical sessions and guided self-study assignments

Assessment Criteria:

End of course unit examination and continuous assessments including a group project

Continuous Assessment 40%	Final Assessment 60%		
Details: Assignments 10 %, Group Project 30 %	Theory (%) 100%	Practical (%) 0%	Other (%)(specify)

Recommended Reading:

1. Joseph Annuzzi Jr., Lauren Darcey, Shane Conder, Introduction to Android Application Development: Android Essentials, 5th Edition, Addison Wesley, 2015
2. Jim Doherty, Wireless and Mobile Device Security, Jones & Bartlett Learning, LLC, 2016
3. Bill Phillips, Chris Stewart & Kristin Marsicano, Android Programming: The Big Nerd Ranch Guide, 3rd Edition, Big Nerd Ranch Guides, 2017
4. Dawn Griffiths, David Griffiths, Head First Android Development, 2nd Edition, O'Reilly, 2017
5. Madona S. Wambua , Modern Android 13 Development Cookbook, Packet Publishing, 2023
6. Material provided in CAL.

Year of Study	2		
Semester	2		
Course Code	INTE 22313		
Course Name	Software Design Patterns and Frameworks		
Pre-requisites	INTE 21213 - Information Systems Modelling		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical/ Tutorial	Self-Guided Study
	45	30	75

Course Aim/Intended Learning Outcomes:

After completing this module, the students should be able to,

1. Identify and explain design principles and patterns in software
2. Apply design principles and patterns when working with software
3. Describe the purpose with and consequences of design patterns
4. Critically evaluate consequences of design patterns with respect to desired software qualities
5. Analyze the relationship between design principles, design patterns, and programming language design, application frameworks, or application domains.
6. Understand trending software frameworks
7. Apply suitable designing patterns and frameworks when software development

Course Content:

Software design, design patterns types (creational, structural, behavioral), history of design patterns, testability, domain-specific languages, dependency injection, SOLID principles, refactoring existing code, trending software frameworks in standalone application development, trending software frameworks in web application development, trending software frameworks in mobile application development, anti-patterns

Teaching /Learning Methods:

Lectures, supervised practical sessions and guided self-study assignments

Assessment Criteria:

End of course unit examination and continuous assessments

Continuous Assessment	Final Assessment		
40 – 50 %	50 – 60 %		
Details: Assignments 40 %	Theory (%) 50%	Practical (%) 50%	Other (%)(specify)

Recommended Reading:

1. Eric Freeman, Elisabeth Robson, Bert Bates, Kathy Sierra, Head First Design Patterns, O'Reilly Media, Inc., 2004
2. Erich Gamma, Richard Helm, Ralph Johns, John Vlissides, Design Patterns: Elements of Reusable Object Oriented Software, 1994
3. Adam Bien , Real World Java EE Patterns-Rethinking Best Practices 2012
4. Gaurav Aroraa, Jeffrey Chilberto, Scott Hanselman, Hands-On Design Patterns with C# and .NET Core, 2019
5. Dinesh Rajput, Rajesh R V, Building Microservices with Spring 2018
6. Materials provided in the CAL.

Year of study 3:

Year of Study	3		
Semester	1		
Course Code	INTE 31233		
Course Name	Human Computer Interaction		
Pre-requisites	INTE 11213 - Fundamentals of Computing		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical/ Tutorial	Self-Guided Study
	45	0	105

On completion of this course, the student should be able to:

1. Understand the classification of input/output devices and techniques, and specifications of human-computer dialogue
2. Describe and evaluate methods of design, prototyping and evaluation of a user interface
3. Explain how different disciplines (human factors, cognitive psychology, engineering, graphics design, etc.) Influence the design, evaluation and implementation of interactive computer systems for human use
4. Develop appropriate interfaces by recognizing how different disciplines (human factors, cognitive psychology, engineering, graphics design, etc,) influence the design, evaluation and implementation of interactive computer systems

Course Content:

Principles of human computer interaction(HCI) and user experience(UX) design, human factors in design, cognitive engineering and human performance: perception psychophysics, attention, time-sharing, workload and its implications on human performance, user psychology, usability engineering, know the user and tasks, usability engineering in modern technologies and devices, usability engineering in application development, usability benchmarking, goal orientated and effective interaction design: storyboarding, prototyping, usability metrics, usability testing and evaluation, usability in practice, universal usability, design issues, data visualization, emerging trends in human computer interaction.

Teaching /Learning Methods:

Lectures, supervised practical sessions and guided self-study assignments

Assessment Criteria:

End of course unit examination and continuous assessments

Continuous Assessment 25 - 50%	Final Assessment 50 - 75%		
Details: quizzes %, mid-term %, Assignments% 0 - 50 %, 0 – 50 % 0 – 50 %	Theory (%) 100%	Practical (%) 0%	Other (%)(specify)

Recommended Reading:

1. Shneiderman, Cohen, M, B., Elmqvist, N., " Designing the User Interface: Strategies for Effective Human-Computer Interaction", 6th Edition, 2021, Pearson
2. Anthony Conta, The Art and Science of UX: A step-by-step guide to designing amazing user experiences, 1st edition, 2023, New Riders
3. Alan J. Dix, Janet E. Finlay, Gregory D. Abowd, Russell Beale, "Human-Computer Interaction", 3rd Edition, 2004, Prentice Hall.
4. Jakob Nielsen, "Usability Engineering", 1993, Academic Press
5. Joel Katz, Designing Information: Human Factors and Common Sense in Information Design, Wiley, 2012
6. Materials provided in the CAL

Year of Study	3		
Semester	1		
Course Code	INTE 31283		
Course Name	Big Data and Data Warehousing		
Pre-requisites	INTE 12223 - Database Design and Development		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	45		105

On completion of this course, the student should be able to:

1. Explain of the theories and principles of data warehousing.
2. Explain the techniques and tools used to design a data warehouse.
3. Design multi-dimensional data structures.
4. Design Data Integration Solutions.
5. Employ HDFS for file storage and retrieval.
6. Implement of in-memory databases.

Course Content:

Data Warehousing as a process, Data Warehouse Design, Physical Design, Data Warehousing Architectures, Data quality and metadata management, Data warehouse technology decisions, Data Warehouse Governance, Data Cleaning And Integration, Key/Value and Document Databases, Map/Reduce and Hadoop, Large-scale Distributed Databases, In-memory Databases, Data Streams Management, Self Service Data Warehouse techniques and Tools, Data Lakes, Reporting in Data Warehouse,

Teaching /Learning Methods:

Lectures, Practices and case discussions

Assessment Criteria:

End-of-course exam, in-class assignments, mini-project, Quizzes

Continuous Assessment 40 - 60%	Final Assessment 40 - 60%		
Details: Assignments% , mini-project%, quizzes% 0 – 20 %, 0 – 20 %, 0 – 20 %	Theory (%) 0-100%	Practical (%) 0-30%	Other (10%)(attendance)

Recommended Reading:

1. Kimball, R, Reeves, L, Ross, M and Thornthwaite, W, (2008), The Data Warehouse Lifecycle Toolkit,2nd Edition, John Wiley & Sons.
2. Matteo, G and Stefano R, Data Warehouse Design: Modern Principles and Methodologies, 1st Edition, McGraw-Hill Osborne Media.
3. Meier A and Kaufmann M, (2019) SQL & NoSQL Databases: Models, Languages, Consistency Options and Architectures for Big Data Management, Springer Vieweg.
4. Kleppmann M, (2017) Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems, O'Reilly Media.
5. Material provided in CAL.

Year of Study	3		
Semester	1		
Course Code	INTE 31356		
Course Name	Software Development Project		
Pre-requisites	All Core Modules from L1 to L3		
Co-requisites	None		
Credit Value	6		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	0	0	300

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Identify a real world problem and user requirements.
2. Evaluate alternative solutions for the problem identified.
3. Select and apply appropriate methodologies.
4. Design a computer system for a selected problem domain utilizing systems analysis and design methodologies.
5. Demonstrate the design of the proposed solution in a technical report and oral presentation.
6. Select appropriate technologies for the development.
7. Perform appropriate testing on the developed computer system.
8. Demonstrate the development of the proposed solution in a technical report and oral presentation.

Course Content:
Students must propose project topics of their own, subject to the approval of suitability by the department, suitable topics could be proposed from the area of interest or based on problems they have experienced. The department may also propose topics based on current requirements and interests of the faculty. Projects will normally aim at the design of a solution for a real world problem and will involve substantial practical work. System development and testing should be carried out and implementable products should be delivered. Computer based solutions should be demonstrated orally and in a written report form.

Teaching /Learning Methods:
Workshops, seminars, one-to-one supervision and feedback from the staff.

Assessment Criteria:
Poster presentation, progress review presentations, final product demonstration and comprehensive project report.

Continuous Assessment
100%

Details: Will be conducted throughout two semesters (Year of study 2 – Semester 2 & Year of study 3 – Semester 1)

Poster Presentation : 5%	Progress Review Presentation I : 25%	Final Presentation : 25%	Code Review : 25%	Project Report : 20%

Recommended Reading:

1. Student guide book for the Software Development Project available at the department
2. Selected project reports.
3. Material provided in CAL.

Year of Study	3		
Semester	1		
Course Code	INTE 31413		
Course Name	Information Technology Infrastructure		
Pre-requisites	INTE 11213 - Fundamentals of Computing INTE 12243 - Computer Networking		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Independent Learning
	40	15	95

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Explain key infrastructure concepts, including how it functions, how to define critical functions, and how to plan and manage infrastructure
2. Explain the components of IT infrastructure solutions
3. Explain the principles of network software and configuration
4. Examine and critique IT infrastructure for organizations designed with security principles
5. Examine and critique IT server architecture (both physical or cloud-based)
6. Explain concepts of Enterprise Architecture

Course Content:
Functions of IT infrastructure; Role of IT infrastructure in business; Components of IT infrastructure; Green infrastructure; Layers of the TCP/IP protocol suite; Network security; Components of Client/server; Wired and wireless networks; Configuration and setup processes on network hardware, software and other supporting devices and components; Network topologies; Software Defined Networks; Network topologies; Security for systems and networks; Infrastructure planning; Continuity planning; Server Components; Server and VM configuration, Cloud configuration; Enterprise architecture models, ITIL

Teaching /Learning Methods:
Lectures, self-study assignments and group work

Assessment Criteria:
End-of- semester examination and continuous assessment

Continuous Assessment 40% - 60%	Final Assessment 40% - 60%		
Details: Individual Assignments, Group Assignments, Quizzes	Theory (%) 100%	Practical (%) 0%	Other

Recommended Reading:

1. Sjaak, L, (2013), IT Infrastructure Architecture: Infrastructure Building Blocks and Concepts, 2nd Edition, Lulu Press Inc.
2. Kirk, H and Susan, L C, (2010), IT Architecture for Dummies, Wiley Publishing.
3. Modern System Administration: Managing Reliable and Sustainable Systems, Jennifer Davis, 2022, 1st Edition, O'Reilly Media
4. Material provided in CAL.

Year of Study	3		
Semester	1		
Course Code	INTE 31243		
Course Name	Software Quality Engineering		
Pre-requisites	INTE 21343 - Software Engineering Concepts		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown	Theory	Practical/ Tutorial	Self-Guided Study
(Notional Hrs.)	45	30	75

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Identify the role and responsibilities of quality engineering
2. Demonstrate a s ound understanding of software quality assurance process
3. Apply different software validation and verification techniques
4. Conduct software inspections, reviews and assess error proneness and testability
5. Design and implement comprehensive test plans
6. Apply various software testing techniques in an effective and efficient manner
7. Determine test coverage and yield according to different criteria
8. Apply different software testing tools

Course Content:
Software quality engineering, software quality assurance process, software validation and verification, software inspections and reviews, software testing principles, test design techniques, different testing types, defect and defect reporting, QA documentation and QA metrics, software error proneness and testability, specification-based testing, code-based testing, alpha, beta and acceptance testing, test prioritisation, coverage criteria, software process and product quality, software quality standards, software testing tools(UI automation, load testing, performance testing, code quality analysis, security testing etc.), statistical techniques to evaluate the defect density and reliability

Teaching /Learning Methods:
Lectures, supervised hands on practical sessions, tutorials and group work

Assessment Criteria:
End-of-semester examination and continuous assessment

Continuous Assessment 20 - 40%	Final Assessment 60 - 80%		
Details: quizzes %, mid-term %, Assignments% 0 - 20 %, 0 – 20 % 0 – 40 %	Theory (%) 100%	Practical (%) 0%	Other (%)(specify)

Recommended Reading:

1. Jeff Tiian (2005), Software Quality Engineering: Testing, Quality Assurance, and Quantifiable Improvement 1st Edition, Wiley-IEEE Computer Society Press,
2. Markus Gartner (2012), ATDD by Example: A Practical Guide to Acceptance Test-Driven Development¸ Addison-Wesley Professional; 1st edition
3. Arnon Axelrod (2018), Complete Guide to Test Automation: Techniques, Practices, and Patterns for Building and Maintaining Effective Software Projects 1st ed. Edition, ,Apress;
4. Geradus Blokdyk(2020), Acceptance Test Driven Development A Complete Guide - Kindle Edition 5STARCooks
5. Materials provided in the CAL

Year of Study	3		
Semester	1		
Course Code:	INTE 31373		
Course Name:	Machine Learning		
Credit Value:	3		
Pre-Requisites	INTE 22303 - Artificial Intelligence		
Co-Requisites	None		
Hourly Breakdown	Theory	Practical	Independent Learning
	100	50	0

Course Aim/Intended Learning Outcomes:
At the completion of this course student will be able to:

1. Describe the machine learning pipeline
2. Differentiate supervised, unsupervised and reinforced machine learning
3. Demonstrate the learning process behind commonly used machine learning algorithms
4. Compare and contrast machine learning algorithms pertaining to a given problem context
5. Apply machine learning algorithms to solve real-world problems
6. Use an appropriate programming language/tool to implement machine learning systems

Course content:
Introduction to machine learning; big data and machine learning applications; the machine learning pipeline; supervised learning – predictive modeling, regression analysis, classification, common machine learning algorithms for predictive modeling, model selection and regularization, handling dimensionality; re-sampling methods – cross validation and bootstrapping; tree-based methods for predictive modeling – decision trees, bagging, random forests, boosting; machine learning for time series prediction; unsupervised learning – introduction to clustering, basic clustering methods; reinforcement learning – model-based and model-free reinforcement learning systems

Teaching /Learning Methods:

A combination of lectures and hands-on practical assignments

Assessment Strategy:

Based on quizzes, group assignments and final examination

Continuous Assessment 25 - 40 %	Final Assessment 60 - 75 %		
Details: Group Assignments 100 %	Theory (%) 0 - 100	Practical (%) 0 - 100	Other (%) (specify) N/A

References/Reading Materials:

1. James, G., Witten, D., Hastie, T., Tibshirani, R., An Introduction to Statistical Learning: With applications in R, Springer, 2017
2. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, 3E, O'Reilly Media, 2022
3. Dangeti, P., Statistics for Machine Learning: Build supervised, unsupervised, and reinforcement learning models using both Python and R, Packt Publishing, 2017
4. Material provided in the CAL

Year of Study	3		
Semester	1		
Course Code	INTE 31393		
Course Name	Information Security		
Pre-requisites	INTE 12243 - Computer Networks INTE 21313 - Business Information Systems		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	40	15	95

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Explain the need for information security and security and risk management
2. Develop an information security Architecture
3. Assess strategies and technologies for asset security, communication and network security, and identity and access management
4. Explain and demonstrate the ability to conduct security Assessments and testing
5. Explain and evaluate information security operations
6. Assess strategies and technologies for software development security

Course Content:
Information security concepts; Security governance principles; Legal and regulatory issues; Investigation types; Security policy, standards, procedures, and guidelines; Personnel security policies and procedures; Threat modelling concepts and methodologies; Risk management; Security awareness, education, and training programmes; Security of information assets; Data security controls and compliance requirements; Secure design principles; Security models; Assess and mitigate the vulnerabilities; cryptographic solutions; Cryptanalytic attacks; Site and facility security; Secure design principles in network architectures; Secure network components; Secure communication channels; Physical and logical access to assets; Identification and authentication; Federated identity with a third-party service; Authorization mechanisms and systems; Security control testing; Security audits; investigations; Configuration Management; Security operations; Resource protection; Incident management; Detective and preventative measures; Patch and vulnerability management; Change management; Recovery strategies; physical security; Personnel safety and security; integrating security in the Software Development Life Cycle; Security impact of acquired software

Teaching /Learning Methods:
Lectures, supervised practical sessions and guided self-study assignments

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment 25 – 50%	Final Assessment 50 – 75%		
Details: quizzes %, mid-term %, Assignments% 25 – 50 %, 0 – 25 % 25 – 50 %	Theory (%) 100%	Practical (%) 0%	Other (%)(specify)

Recommended Reading:

1. Principles of Information Security, 2022, Michael E. Whitman and Herbert J. Mattord, 7th Edition, Cengage Learning
2. The Official (ISC)2 CISSP CBK Reference, 2021, Arthur J. Deane and Aaron Kraus, 6th Edition, John Wiley
3. Information Security: Principles and Practice, 2021, Mark Stamp, 3rd Edition, John Wiley
4. Fundamentals of Information Systems Security, 2021, David Kim and Michael G. Solomon, 4th Edition, Jones & Bartlett Learning
5. Designing Secure Software: A Guide for Developers; 2021; Loren Kohnfelder; No Starch Press
6. Material provided in LMS.

Year of Study	3		
Semester	1		
Course Code	INTE 31403		
Course Name	System Administration and Maintenance		
Pre-requisites	INTE 11213 - Fundamentals of Computing INTE 12243 - Computer Networks		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	30	45	75

Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Explain system administration and how it facilitates operations in an organization.
2. Use industry-level tools and techniques to plan, design, and implement system solutions satisfying organizational requirements.
3. Reflect on how hardware and software systems are implemented and apply that knowledge in analysing real-world issues related to system performance, scalability, and security.

Course Content:
Introduction to system administration, Job descriptions for system administrators, UNIX history and basics, documentation techniques, storage models and disks, devices and interfaces, storage virtualization, Filesystem basics, boot process and MBR, software types, software installation, package management, networking, DNS, HTTP, HTTPS, ICMP, ARP, NDP, e-mail, automation, regex, scripting, monitoring, backup and disaster recovery, configuration management, system security, risk assessment, threat models, ethics and social responsibilities.

Teaching /Learning Methods:
Lectures, supervised practical sessions, and group work.

Assessment Criteria:
End-of-semester examination and continuous assessment.

Continuous Assessment:	Semester-End Examination:		
40 - 60%	40 - 60%		
Details: Mid-term and end-term quizzes: 20 – 30% Assessments: 20 – 30%	Theory (%) 40 – 50%	Practical (%) 20 – 30%	Other (%) N/A

Recommended Reading:

1. Mark Burgess, 2004, Principles of Network and System Administration, 2nd edition, Wiley.
2. Thomas A. Limoncelli, Christine Hogan, Strata R. Chalup, 2016, Practice of System and Network Administration - DevOps and other Best Practices for Enterprise IT - Volume 1, 3rd edition, Addison-Wesley Professional.
3. Thomas A. Limoncelli, Christine Hogan, Strata R. Chalup, 2014, The Practice of Cloud System Administration - DevOps and SRE Practices for Web Services - Volume 2, 1st Edition, Addison-Wesley Professional.
4. Materials provided in the CAL.

Year of Study	3		
Semester	1		
Course Code:	INTE 31423		
Course Name:	Data Analytics and Visualization		
Credit Value:	3		
Pre-Requisites	INTE 21313 - Business Information Systems		
Co-Requisites	None		
Hourly Breakdown	Theory	Practical	Independent Learning
	100	25	25

Course Aim/Intended Learning Outcomes:
At the completion of this course students will be able to:

1. Identify data as a strategic resource
2. Explain the importance of data analytics in organisations
3. Describe the stages of an industry-level data analytics process framework
4. Explain different types of data analytics
5. Design visualisations to understand and interpret data
6. Use data analytics and visualisation to solve a real-world problem

Course Content:
Introduction to big data and analytics; The data analytics process; Types of data; Data extraction and understanding; Data visualization; Data preprocessing; Types of data analytics – descriptive, diagnostic, predictive, and prescriptive; Role of artificial intelligence and machine learning in data analytics; Descriptive analytics techniques – clustering and association rule mining techniques; Predictive analytics techniques – classification and regression analysis techniques; Prescriptive analytics techniques; Responsible use of data analytics

Teaching /Learning Methods:
Lectures, Practical sessions using common no-code tools, and Group Assignments

Assessment Strategy:

Continuous Assessment 25-40 %	Final Assessment 60-75 %		
Details: Quizzes 25%, Assignment 75 %	Theory (%) 0-100	Practical (%) 0-100	Other (%) (specify) N/A

References/Reading Materials:

1. Elizabeth Clarke, Everything Data Analytics A Beginner's Guide to Data Literacy: Understanding the Processes That Turn Data Into Insights, Ind. publisher, 2022
2. Cole Nussbaumer Knaflic, Storytelling with Data: A Data Visualization Guide for Business Professionals, 1E, Wiley, 2015
3. Materials provided in the CAL

Year of study 4:

Year of Study	4		
Semester	1		
Course Code	INTE 41403		
Course Name	Advanced Databases		
Pre-requisites	INTE 12223 - Database Design and Development		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown	Theory	Practical	Guided Self-study
(Notional Hrs.)	30	45	75

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Explain the concepts of transaction processing in Database Management Systems.
2. Select mechanisms for dealing with transaction management and security of database systems.
3. Explain database recovery mechanisms.
4. Evaluate different database systems.

Course Content:
Relational algebra, relational calculus, query optimization, concurrency control, database recovery, database security, distributed databases, parallel databases, NoSQL databases, Cloud Databases, Data Migration Strategies, Master Data Management (MDM) system

Teaching /Learning Methods:
Lectures, supervised practical sessions and group work

Assessment Criteria:
End-of-semester examination, assignments and quizzes

Continuous Assessment 40 - 60%	Final Assessment 40 - 60%		
Details: Assignments% , mini-project%, quizzes% 0 – 40 %, 0 – 20 %	Theory (%) 50%	Practical (%) 50%	Other (10%)(attendance)

Recommended Reading:

1. Hector Garcia-Molina, Jeff Ullman, and Jennifer Widom, (2009), Database Systems: The Complete Book, 2nd Edition, Pearson.
2. Ramez Elmasri and Shamkant B. Navathe, (2015), Fundamentals of Database Systems, 7th Edition, Pearson.
3. Wilfried Lemahieu, Seppe vanden Broucke, Bart Baesens, (2018), Principles of Database Management: Practical Guide to Storing, Managing and Analyzing Big and Small Data, 1st edition, Cambridge University Press.
4. Thomas M. Connolly, Carolyn Begg, (2015), Database Systems: practical approach to design, implementation, and management, 6th edition, Pearson Education Limited.
5. C. J. Date, (2003), An Introduction to Database Systems, 8th Edition, Pearson.
6. Material provided in CAL**.**

Year of Study	4		
Semester	1		
Course Code	INTE 41413		
Course Name	Internet of Things		
Pre-requisites	INTE 12243 - Computer Networks INTE 22263 - Embedded Systems Development		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown	Theory	Practical	Guided Self-study
(Notional Hrs.)	30	45	75

Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Explain layers of abstraction in IoT architecture, and how these facilitate development of IoT applications.
2. Use industry-level tools and techniques to plan, design, simulate, and implement sustainable IoT solutions.
3. Reflect on how hardware and software components, and protocols are implemented and apply that knowledge in analysing issues related to IoT application performance, scalability, and security.

Course Content:
Overview of IoT, sensors, actuators, connectivity, people and processes, IoT and Internet, IoT architectures and protocols, IoT building blocks, IoT communication protocols, security in IoT, IoT value modelling, IoT platforms, IoT applications and business use-cases.

Teaching /Learning Methods: s
Lectures, supervised practical sessions, and group work.

Assessment Criteria:
End-of-semester examination, projects, and continuous assessment.

Continuous Assessment: 40 - 60%	Semester-End Examination: 40 - 60%		
Details: Mid-term and end-term quizzes: 20 – 30% Assessments: 20 – 30%	Theory (%) 40 – 50%	Projects (%) 20 – 30%	Other (%) N/A

Recommended Reading:

1. Giacomo Veneri, Antonio Capasso, 2018, Hands-On Industrial Internet of Things: Create Powerful Industrial IoT Infrastructure using Industry 4.0, Packt Publishing.
2. Marco Schwartz, 2016, Internet of Things with Arduino Cookbook, Packt Publishing.
3. Marco Schwartz, 2016, Internet of Things with ESP8266, Packt Publishing.
4. John Davies, Carolina Fortuna, The Internet of Things – From Data to Insights, 2020, Wiley.
5. Brian Russell, Drew Van Duren, 2016, Practical Internet of Things Security, Packt Publishing.
6. Materials provided in the CAL.

Year of Study	4		
Semester	1		
Course Code	INTE 41393		
Course Name	System Integration Technologies		
Pre-requisites	INTE 12213 - Object Oriented Programming, INTE 21323 - Web Applications Development, INTE 22313 - Software Design Patterns and Frameworks		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown	Theory	Practical/ Tutorial	Self-Guided Study
(Notional Hrs.)	45	0	105

Course Aim/Intended Learning Outcomes:

After completing this module, the students should be able to,

1. Explain how integrating various modules for produce a working computer system.
2. Describe how integration as an important function of all it professionals.
3. Understand tools and techniques required for devops
4. Describe how to deploy devops more efficiently, avoid problems, and drive far more value
5. Evaluate and apply different technologies that organizations typically use integrating computer systems

Course Content:

Importance of integration for IT professionals, Architectures for integrating systems, Data mapping and exchange, Web services for system integration, Socket programming(TCP/IP, Datagram), Message and queuing services, Low level data communication, Metadata, Data representation and encoding, Program paradigms (Imperative, Procedural, Declarative, Functional, Logical, Object Oriented, Event-driven), Scripting techniques(Operating system, Shell, Bash), Compilers and interpreters, Evidence-based security, Code access security, Encryption of data between systems and services, Secure coding and secure coding practices (Preventing buffer overflow, securing state data, Securing method access, Wrapper code, Unmanaged code, Validation of user input, Remoting considerations, Protected objects, Serialization, Robust error handling), Authentication, Versioning and version control, DevOps practices and their implications on system design, DevOps and Software Development Life Cycle, Team practices related to DevOps, Cloud computing- What is Cloud? Evolution of Cloud Computing, IAAS (Infrastructure as a Service), SAAS (Software as a Service), PAAS (Platform as a Service), Private, Public and Hybrid Cloud, Continuous Integration and Deployment

Teaching /Learning Methods:

Lectures, supervised practical sessions and guided self-study assignments

Assessment Criteria:

End of course unit examination and continuous assessments

Continuous Assessment 25 - 50%	Final Assessment 50 - 75%		
Details: quizzes %, mid-term %, Assignments% 0 - 50 %, 0 – 50 % 0 – 50 %	Theory (%) 100%	Practical (%) 0%	Other (%)(specify)

Recommended Reading:

1. Bass, L., Weber, I., Zhu, I., (2015), DevOps: A Software Architect's Perspective, |Addison-Wesley Professional
2. Limoncelli, T. A., Hogan, C. J., Chalup, S. R., (2017), Practice of System and Network Administration, The: Volume 1: DevOps and other Best Practices for Enterprise IT, 3rd Edition, Addison-Wesley Professional
3. Material provided in CAL

Year of Study	4		
Semester	1 OR 2		
Course Code	INTE 44303		
Course Name	Information Audit and Assurance		
Pre-requisites	INTE 21313 - Business Information Systems INTE 31393 - Information Security INTE 12243 - Computer Networks		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	40	15	95

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Plan an audit to determine whether information systems are protected, controlled, and provide value to the organization.
2. Evaluate the IT strategy, governance and operations for alignment with the organization's strategies and objectives
3. Evaluate whether the organization's objectives are being met by the management practices of its information systems
4. Evaluate the organization's ability to continue business operations
5. Evaluate whether information asset protection and security event management mechanisms are appropriate

Course Content:
Introduction to IS audits; Types of Controls; Risk-Based Audit Planning; Audit Evidence Collection Techniques; Analytics and reporting; Evaluation of business needs and rationale for information systems acquisition; Evaluation of development, testing, deployment and migration strategies used; Evaluation of management of the acquisition project; Evaluation of Information Technology governance, operations, incident management; Assessing the effectiveness of common technology components, IT Asset Management, operations, systems' performance, data governance, service levels, business continuity; Evaluation of change, configuration, release, and patch management; Evaluation of information asset security, adherent to privacy principles, access to assets, identity and access management; Assessing the effectiveness of network and end-point security, data Encryption, communication techniques, virtualized Environments, mobile, wireless, and Internet-of-Things (IoT) devices; Evaluation of security awareness training and programs. Analyzing the attack methods and techniques; Assessing the effectiveness of security testing tools and techniques, security monitoring tools and techniques, and evidence collection and forensics
Information security management frameworks; Information security policies, standards and guidelines; Information security planning

Teaching /Learning Methods:
Lectures, supervised hands-on practical sessions, and group work

Assessment Criteria:
End-of-semester examination and continuous assessment

Continuous Assessment 40 – 60%	Semester-End Examination 60 – 40%		
Details: Assignments: 50% MCQs: 50%	Theory (%) 100%	Practical (%)	Other (%)(specify)

Recommended Reading:

1. Information Technology Auditing, 2015, James A. Hall, 4th Edition, Cengage Learning
2. Information Technology Control and Audit, Angel R. Otero, 2019, Fifth Edition, Taylor and Francis
3. Certified Information Systems Auditor® ISACA manual
4. Materials provided in the LMS.

Year of Study	4		
Semester	1		
Course Code	INTE 41283		
Course Name	Information Systems Management and Strategy		
Pre-requisites	INTE 21313 – Business Information Systems		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	45	-	105

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Explain how information systems/ information technology provide a key source of competitive advantage for organizations
2. Explain the concepts of information economics at the enterprise level
3. Structure IS-related activities to maximize the business value of IS within and outside the company
4. Develop information systems/ information technology strategic plans
5. Examine how strategic decisions are made concerning acquiring IS resources and capabilities
6. Evaluate the role of IT control and service management frameworks for effective management of information technology

Course Content:
The IS function, IS strategic alignment, strategic use of information, impact of IS on organizational structure and processes, role of IS in defining and shaping competition, Porter's models, IS economics, managing the information systems function, financing and evaluating the performance of information technology investments, acquiring information technology resources and capabilities, using IS/IT governance frameworks

Teaching /Learning Methods:
Lectures and guided self-study assignments

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment	Final Assessment		
25 – 50%	50 – 75%		
Details: quizzes %, mid-term %, Assignments% 0 – 25 %, 0 – 25 % 25 – 50 %	Theory (%) 100%	Practical (%) 0%	Other (%)(specify)

Recommended Reading:

1. Managing and Using Information Systems: A Strategic Approach, 2024, Keri E. Pearlson, Carol S. Saunders, Dennis F. Galletta, 8th Edition, Pearson Higher education, John Wiley
2. Joe Peppard, John Ward, The Strategic Management of Information Systems: Building a Digital Strategy, 4th Edition, John Wiley, 2016
3. Material provided in CAL

Year of Study	4		
Semester	1		
Course Code	MGTE 41363		
Course Name	Digital Innovations Management		
Pre-requisites	INTE 21313 - Business Information Systems		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	45	-	105

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Examine the unique features that an application of emerging technology may offer
2. Evaluate the role of digital business technologies in social and mobile domains
3. Identify and assess characteristics necessary for digital innovation
4. Identify and validate an opportunity to develop a new digital business model
5. Identify and evaluate key issues related to implementation and infrastructure issues

Course Content:
Emerging digital technology areas; Characteristics of new digital technologies; Social, mobile and business applications of technology; Conditions, domain and applications for digital innovation; Digital business models; Digital business innovation elements; Ideation for a new business model; Implementation criteria innovative technologies; Infrastructure elements for implementation; Key criteria for continuing or stopping implementation; Managing innovative projects; Evaluation criteria for a successful application

Teaching /Learning Methods:
Lectures, supervised practical sessions and guided self-study assignments

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment 25 – 50%	Final Assessment 50 – 75%		
Details: quizzes %, mid-term %, Assignments% 25 – 50 %, 0 – 25 % 25 – 50 %	Theory (%) 100%	Practical (%) 0%	Other (%)(specify)

Recommended Reading:

1. Handbook of Digital Innovation, 2020, Edited by Satish Nambisan, Nancy and Joseph Keithley, Kalle Lyytinen, Youngjin Yoo, Edward Elgar publishing
2. Handbook of Social Computing, 2024, Edited by Peter A. Gloor, Francesca Grippa, Andrea Fronzetti Colladon, Aleksandra Przegalinska, Edward Elgar publishing
3. Competitive Digital Innovation: Transforming Products, Processes and Business Models to Win in the Digital Economy, Amit Basu and Steve Muylle, 1st edition, Palgrave Macmillan
4. Demand-Driven Business Strategy: Digital Transformation and Business Model Innovation, 2022, Cor Molenaar, 1st Edition, Routledge.
5. Material provided in CAL.

Year of Study	4		
Semester	1		
Course Code	INTE 41373		
Course Name	Image Processing		
Pre-requisites	INTE 11213 - Fundamentals of Computing, INTE 22343 - Data Structures and Algorithms, PMAT 31212 - Mathematics for Computing - 3		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown	Theory	Practical/Tutorial	Self-Guided Study
(Notional Hrs.)	30	45	25

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Describe the fundamental properties of digital images
2. Explain the mathematical representations of digital images
3. Describe concepts and approaches used in image processing
4. Design and implement algorithms for performing basic image processing
5. Design and implement algorithms for advanced image analysis
6. Apply image processing in solving real world problems

Course Content:
Image representation, properties of digital images, image formats, sampling and quantization, relationship between pixels, image enhancement in spatial domain, grey level transformation, histogram-based techniques, arithmetic/logical operations, spatial filtering, noise removal, image enhancement in frequency domain, image degradation and noise, , image restoration, geometric transformation, edge detection, image segmentation, morphology, image compression, colour image processing

Teaching /Learning Methods:
Lectures, supervised hands on practical sessions, and guided self-study assignments

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment	Final Assessment		
0 - 25%	75 - 100%		
Details: quizzes, mid-term, assignments 0 - 25 %, 0 – 25 %, 0 – 25 %	Theory (%) 50%	Practical (%) 50%	Other (%) (specify)

Recommended Reading:

1. Rafael C. Gonzalez and Richard E. Woods (2017), Digital Image Processing, 4th edition, Pearson
2. Wilhelm Burger and Mark J. Burge (2022), Digital Image Processing: An Algorithmic Introduction, 3rd edition, Springer.
3. Materials provided in the CAL

Year of Study	4		
Semester	1		
Course Code:	INTE 41323		
Course Name:	Neural Networks and Deep Learning		
Credit Value:	3		
Pre-requisites	INTE 22303 - Artificial Intelligence		
Co-requisites	None		
Hourly Breakdown	Theory	Practical	Independent Learning
	100	25	25

Course Aim/Intended Learning Outcomes:
At the completion of this course student will be able to:

1. Describe the basic steps of neural network learning
2. Explain the difference between traditional neural networks and deep neural networks
3. Evaluate deep learning algorithms with respect to a given problem
4. Implement deep learning algorithms using state of the art software libraries
5. Discuss emerging deep neural network architectures

Course Content:
Introduction to neural networks – basic architecture, backpropagation learning, practical problems of neural networks, non-linear activation functions, common neural architectures; Applications of shallow neural networks; Training deep neural networks – backpropagation, setup and initialization issues, vanishing and exploding gradient problems, gradient descent strategies; Generalizing deep neural networks; Radial Basis Function Networks; Restricted Boltzmann Machines; Recurrent Neural Networks; Convolutional Neural Networks; Generative AI and deep learning; Transformers and other Emerging topics in deep neural networks

Teaching /Learning Methods:

Lectures, Seminars, Group Assignments and Practical Sessions

Assessment Strategy:

Continuous Assessment 50 %	Final Assessment 50 %		
Details: Quizzes 25%, Assignments 75 %	Theory (%) 100%	Practical (%) 0%	Other (%) (specify) N/A

References/Reading Materials:

1. Aggarwal, C. C., Neural Networks and Deep Learning – A Textbook, 1E, Springer, 2018
2. Géron, A., Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, 2E, O'Reilly Media, 2019
3. David Foster, Generative Deep Learning: Teaching Machines To Paint, Write, Compose, and Play, 2E, O'Reilly Media, 2023
4. Materials provided in CAL

Year of Study	4		
Semester	1		
Course Code	INTE 41383		
Course Name	Industrial Automation		
Pre-requisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown	Theory	Practical/ Tutorial	Self-Guided Study
(Notional Hrs.)	45	15	90

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Identify And Evaluate Industrial Automation Requirements
2. Select And Apply Appropriate Sensors And Actuators
3. Design And Program Plc Systems
4. Design, Program And Connect Plc Systems For Iot Applications

Course Content:
Industrial Automation: Introduction to Industrial Automation, requirements and constraints, sensors and transducers, actuating devices, signal conditioning and real time interfacing, elements of a data acquisition and control system, transducers and signal conditioning, devices for data conversion, data conversion process
PLC Systems: discrete input/output system, analog input/output system, serial function input/output and serial communication interfacing, PLC programming languages, timers, counters, system programming and implementation, PLC system documentation, IoT for Industry 4.0

Teaching /Learning Methods:
Lectures, supervised practical sessions and guided self-study assignments

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment 25 – 50%	Final Assessment 50 – 75%		
Details: quizzes %, mid-term %, Assignments% 0 – 50 %, 0 – 50 % 0 – 50 %	Theory (%) 100%	Practical (%) 0%	Other (%)(specify)

Recommended Reading:

1. Manesis, S., & Nikolakopoulos, G. (2018). Introduction to Industrial Automation. CRC Press 2018
2. Mehta, B. R., & Reddy, Y. J. (2014). Industrial process automation systems: Design and implementation. Elsevier Inc 2014
3. Mikell Groover (2019) Automation, Production Systems, and Computer-Integrated Manufacturing, 5th Edition, Pearson Publications 2019
4. Materials provided in CAL

Year of Study	4		
Semester	2		
Course Code	INTE 42353		
Course Name	Semantic Web and Ontological Engineering		
Pre-requisites	INTE 12213 - Object Oriented Programming, INTE 21323 - Web Applications Development		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown	Theory	Practical/ Tutorial	Self-Guided Study
(Notional Hrs.)	45	15	90

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Describe The Core Standards And Technical Components Of The Semantic Web And The Concept Of Ontology.
2. Design And Develop Ontology Models For Semantic Web Applications.
3. Describe The Trends In Semantic Web

Course Content:
Introduction to traditional web vs semantic web, Resource Description Framework (RDF), Other RDF related technologies (RDFa, Microformats, GRDDLE), RDFS and Ontology, Web Ontology Language (OWL), Querying the semantic web-SPARQL, Semantic web real world examples and applications, Development tools for the semantic web, Data Science and Ontologies, Web mining and ontology.

Teaching /Learning Methods:
Lectures, supervised practical sessions and guided self-study assignments

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment 25 – 50%	Final Assessment 50 – 75%		
Details: quizzes %, mid-term %, Assignments% 0 – 50 %, 0 – 50 % 0 – 50 %	Theory (%) 100%	Practical (%) 0%	Other (%)(specify)

Recommended Reading:

1. Yu, L (2011), A Developer's Guide to the Semantic Web, Springer-Verlag.
2. Archana Patel (Editor), Narayan C. Debnath (Editor), Bharat Bhushan (Editor), Semantic Web Technologies: Research and Applications, 2022
3. Jack Grealish, Driven Ontology Web Mining Paperback, 2023, Shine Publisher
4. Materials provided in the CAL

Year of Study	4		
Semester	2		
Course Code	INTE 42333		
Course Name	Complex Systems and Agent-Based Modeling		
Pre-requisites	INTE 11223 - Programming Concepts, INTE 22303 - Artificial Intelligence		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	50	50	50

Course Aim/Intended Learning Outcomes:

After completing this module, the students should be able to,

1. Recognize The Complexity In Real-World Systems
2. Discuss The Evolutionary And Adaptive Nature Of Complex Systems
3. Appreciate Agent-Based Modeling As A 'Third Way' Of Doing Science
4. Use Agent-Based And Artificial Intelligence Techniques To Model Complex Systems
5. Implement Agent-Based Simulation Models To Understand Complex Systems' Dynamics
6. Apply Abm Techniques To Solve Practical Problems In Different Domains

Course Content:

Introduction to complex systems; The concept of 'emergence'; Simulation – A third way of doing science; Agent-Based Modeling (ABM); Creating agent-based simulation models – The KISS principle, Cellular Automata, Network models, Geospatial models, Organizational models; Modeling evolutionary systems; Adaptive systems – Using game theory and repeated games; Machine learning and ABM – Reinforcement Learning, Bayesian Learning, Learning Classifier Systems; Documentation standards and protocols of ABMs; ABM applications; Advancements of ABM research

Teaching /Learning Methods:

Lectures, Practical sessions, Research discussions and Guided self-study assignments

Assessment Criteria:

End of course unit examination and continuous assessments

Continuous Assessment 25-50%	Final Assessment 50-75%		
Details: quizzes %, Assignments%, Project% 0 %, 100 % 0 %	Theory (%) 0-100%	Practical (%) 0-100%	Other (%)(specify)

Recommended Reading:

1. Wilensky, U., Rand, W., An Introduction to Agent-Based Modeling: Modeling Natural, Social, and Engineered Complex Systems with NetLogo, 1E, MIT Press, 2015
2. Railsback, S. F., Grimm, V., Agent-Based and Individual-Based Modeling: A Practical Introduction, 2E Princeton University Press, 2019
3. North, M. J., Macal, C. M., Managing Business Complexity: Discovering Strategic Solutions with Agent-Based Modeling and Simulation, Oxford University Press, 2007
4. Juan Martin Garcia, Agent-Based Modeling and Simulation I & II, Ind. Publisher, 2021
5. Materials made available through CAL

Year of Study	4		
Semester	2		
Course Code:	INTE 42343		
Course Name:	Natural Language Processing		
Credit Value:	3		
Pre-Requisites	None		
Co-Requisites	None		
Hourly Breakdown	Theory	Practical	Independent Learning
	100	25	25

Course Aim/Intended Learning Outcomes:
At the completion of this course student will be able to:

1. Evaluate Text retrieval techniques
2. Develop text analytics systems using state of the art technologies
3. Explain the foundation technologies of question and answering systems
4. Apply emerging generative AI and large language model-based technologies for common NLP tasks
5. Use industry standard tools and text data sets.

Course Content:
Introduction to NLP, Text Retrieval, Text Classification, Word Vectors, Sentiment Analysis, Language modeling and sequence tagging, Parts of Speech & Pos Tagging, Vector Space Models of Semantics, Contact Free Grammar (CFG) & Probabilistic Contact Free Grammar (PCFG), Latent Dirichlet Allocation (LDA), Dialog systems, Neural networks for natural language processing, Transformers and large language models

Teaching /Learning Methods:
Lectures, Quizzes and Group Assignments

Assessment Strategy:

Continuous Assessment 25-40 %	Final Assessment 60-75 %		
Details: Quizzes 25%, Assignment 75 %	Theory (%) 0-100	Practical (%) 0-100	Other (%) (specify) N/A

References/Reading Materials:

1. Jacob Eisenstein , Introduction to Natural Language Processing (Adaptive Computation and Machine Learning series), The MIT Press, 2019
2. Hobson Lane, Hannes Hapke & Cole Howard, Natural Language Processing in Action: Understanding, analyzing, and generating text with Python, Manning Publications, 2019
3. Materials provided in the CAL

b) Management and Technology

Year of study 1:

Year of Study	1		
Semester	1		
Course Code	MGTE 11233		
Course Name	Business Statistics and Economics		
Pre-requisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	45	-	105

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Apply a variety of methods for exploring, summarizing and presenting data.
2. Understand the fundamentals of probability concepts and develop and implement statistical models
3. Describe the concept of scarcity and how markets allocate resources to address basic economic problems
4. Use demand and supply to analyse changes in prices and quantities
5. Illustrate how firms arrive at production decisions based on cost and revenue
6. Classify firms into different market structures and show how competitive structures influence the individual firm's behaviour
7. Explain the objectives of government macroeconomic policy and describe how they can be pursued through fiscal and monetary policy

Course Content:
Descriptive statistics: compilation, classification, tabulation and diagrammatic and graphical representation of various types of statistical data, skewness and kurtosis, frequency distributions, and measures of location and dispersion.

Elements of probability theory: concepts of probability, sample space, field of events and generalized addition theorem, conditional probability, independence, Bayes' theorem, random variables, distribution theory, expectation, variance, Poisson Distribution, and Normal Distribution
The concepts of scarcity and opportunity cost; production possibility frontier; Economic systems; Demand and supply; Elasticity; Government controls; Utility theory; Supply decisions; Market structures; Introduction to macroeconomics

Teaching /Learning Methods:
Lectures, supervised practical sessions and guided self-study assignments, interactive classroom sessions, and applications of Microsoft Excel

Assessment Criteria:
End of course unit examination, continuous assessments, tutorials, and class attendance

Continuous Assessment 25 – 50%	Final Assessment 50 – 75%		
Details: quizzes %, mid-term %, Assignments% 25 – 50 %, 0 – 25 % 25 – 50 %	Theory (%) 100%	Practical (%) 0%	Other (%)(specify)

Recommended Reading:

1. Levine, D M, Stephan, D, Krehbiel, T C and Berenson, M L, (2011). Statistics for managers: using Microsoft Excel.Prentice Hall.
2. Lind, D, Marchal, W G and Wathen, S, (2011). Statistical Techniques in Business and Economics. McGraw-Hill.
3. Anderson, D R, Sweeney, D J and Williams, T A (2010). Statistics for Business and Economics. South-Western College Pub.
4. Steven A. Greenlaw and David Shapiro, 2022, Principles of Economics, 3rd Edition, openstax.org
5. Dean Karlan and Jonathan Morduch, 2021, Economics, 3rd Edition, McGraw-Hill
6. John Sloman, Dean Garratt, and Jon Guest, 2021, Economics, 11th Edition, Pearson Higher education.
7. Material provided in the eKel

Year of Study	1		
Semester	1		
Course Code	MGTE 11243		
Course Name	Principles of Management & Organizational Behaviour		
Pre-requisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical/ Tutorial	Self-Guided Study
	45	30	75

Course Aim/Intended Learning Outcomes:

After completing this module, the students should be able to,

1. Define the basic management concepts and apply them to practical scenarios
2. Develop a solid understanding of the development of management thought and its evolution
3. Develop a firm knowledge base on the four management functions and their applications in contemporary organisations
4. Identify different contemporary management tools and reasons why management trends change over time.
5. Explain managers' strategies to help organisations adapt to an uncertain or turbulent environment.
6. Identify and develop key managerial competencies to become effective managers.
7. Identify and clarify the sources and implications of emotions and stress at work.
8. Identify perceptions, decision-making, and diversity in organisations.

Course Content:

Understanding organisations, Evolution of management thought, Organization and Environment, Managerial process: Planning and Strategic decision-making, Organizing, Organizational structure and design, Leading and Controlling, Motivation, Motives of Group and Team Behaviour, Organizational communication, Drivers of individual Behaviour, Organizational Culture, Learning and the Processes of Organizational Change, Emotions and Stress at Work, Job attitudes, Perceptions, Decision-Making, and Diversity in Organisations, Perceptions, Decision-Making, and Diversity in Organisations.

Teaching /Learning Methods:

Lectures, in-class discussions, real world case studies and self-study

Assessment Criteria:

End of course unit examination and continuous assessments

Continuous Assessment 25 – 50%	Final Assessment 50 - 75%		
Details: quizzes %, mid-term %, Assignments% 0 - 30 %, 0 – 30 % 0 – 50 %	Theory (%) 10%- 90%	Practical (%) 0%	Other (%)(Attendance) 0%- 10%

Recommended Reading:

1. R. L. Daft, (2016), New Era of Management, 11E, Cenage Learning.
2. S. P. Robins and M. Coulter, (2017), Management, 14E, Pearson.
3. S. P. Robbins, M. Coulter, D. A. De Cenzo, (2016), Fundamentals of Management, 10E, Pearson.
4. S. Certo, (2014), Modern Management, Prentice Hall, Delhi.
5. S. P. Robbins and T. A. Judge, (2012), Organizational Behaviour, Prentice Hall, New Jersey.
6. Robbins, S., Judge, T., Edwards, M., Sandiford, P. and Fitzgerald, M. (2019). Organisational Behaviour. (9th ed.), Pearson Education / Prentice Hall.
7. Materials provided in the CAL.

Year of Study	1		
Semester	2		
Course Code	MGTE 12253		
Course Name	Accounting Concepts and Costing		
Pre-requisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical/ Tutorial	Self-Guided Study
	45	15	90

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Discuss and explain the nature and purpose of accounting
2. Record economic transactions and develop the key financial statements for the enterprise
3. Demonstrate an understanding of the principles of financial and managerial accounting
4. Identify, analyze and evaluate financial information relevant to the operating, investing and financing decisions of a business entity
5. Organize, calculate and analyze information relevant to decision making
6. Identify cost components, cost behaviour and prepare a cost sheet for a product

Course Content:
Concepts of financial accounting, Practice of book keeping including books of Prime Entry, Ledgers and Trial Balance, Preparation of Trading Profit and Loss and Balance Sheet (Income Statement), Preparation of Balanced Sheet (Statement of Financial Position), Cash Flow Statement, Bank Reconciliation Statement, Concepts of costing and preparation of costing sheet.

Teaching /Learning Methods:
Lectures, in-class discussions, real world case studies and self-study

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment 25 – 50%	Final Assessment 50 - 75%		
Details: quizzes %, mid-term %, Assignments% 0 - 30 %, 0 – 30 % 0 – 50 %	Theory (%) 10%- 90%	Practical (%) 0%	Other (%)(Attendance) 0%- 10%

Recommended Reading:

1. Williams, J, Haka, S, Bettner, M andCarcello, J, (2009), Financial Accounting, 14th Edition, McGraw-Hill/Irwin.
2. Hilton, R W, (2011), Managerial Accounting: Creating Value in a Dynamic Business Environment,9th Edition, McGraw-Hill/Irwin.
3. Materials provided in the CAL.

Year of Study	1		
Semester	2		
Course Code	MGTE 12263		
Course Name	Optimization Methods in Management Science		
Pre-requisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical/ Tutorial	Self-Guided Study
	30	45	75

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Identify and analyse business problems and formulate decision problems as mathematical models and lay out as spreadsheet models
2. Assess the main issues in a situation and choose an appropriate approach to tackle them.
3. Apply appropriate quantitative techniques and perform sensitivity analysis in managerial decision-making.
4. Apply tools available in Microsoft Excel to solve complicated problems and generate solutions.
5. Communicate the analysis and results of business decision problems.

Course Content:
Linear Programming: Introduction to Operations Research and Linear Programming (LP), Applications of LP; Problem formulation, Solving LP problems: 1. Graphically, 2. Using the simplex method, and 3. Using Excel Solver, Sensitivity analysis
Integer Programming: Types of integer linear programming models, Pure integer, Mixed integer, Binary, Optimality

Special types of LP Problems: Transportation problems, Transhipment problems, Assignment problems

Network Modelling: techniques for various classes of network problems: minimal-cost maximal flow and shortest path problems

Teaching /Learning Methods:
Lectures, interactive classroom sessions, case discussions, and applications of Microsoft Excel Solver

Assessment Criteria:
End of course unit examination, group assignments, mid-term examination, practical test, presentations, class attendance

Continuous Assessment 40% - 50%	Final Assessment 50% - 60%		
Details: quizzes 0-30%, practical test 0-30%	Theory (%) 0-80%	Practical %) 0-30%	Other (attendance) 10%

Recommended Reading:

1. Anderson, D R, Williams, T A and Sweeney, S J, (2011), An Introduction to Management Science: Quantitative Approaches to Decision Making, 13th Edition, South Western College Pub.
2. Fedrick, S. Hiller, G. Lieberman, B. Nag (2014), Introduction to operations research, McGraw-Hill Professional, 9th Edition, McGraw Hill Education (India) Pvt. Ltd.
3. Barry, R and Ralph, M S (2015), Quantitative analysis for Management 12th Edition Prentice Hall..
4. Kenneth R. Baker (2015) Optimization Modeling with Spread sheets,3rd Edition, Thompson learning academic Resource Center.
5. Materials provided in the eKel.

Year of Study	1		
Semester	2		
Course Code	MGTE 12273		
Course Name	Industry and Technology		
Pre-requisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Class-based case discussions/ Field visits/ Tutorials	Self-Guided Study
	45	30	75

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Describe the general industrial environment in Sri Lanka
2. Define the role of technology in micro enterprises and regional (macro) growth
3. Review technology trends in the past and identify emerging trends
4. Describe the basic manufacturing processes used in industry,
5. Describe basic elements of a manufacturing system

Course Content:
Industrial environment: external and internal factors, Role of technology in industrial development, research and development, invention and innovation
Technology: evolution of technology, technological development, technology and society, product design concepts, manufacturing technologies, principles of engineering, basic manufacturing processes, principles of product and process design, manufacturing systems, sustainable production.

Teaching /Learning Methods:
Lectures, interactive classroom sessions and case analysis

Assessment Criteria:
End-of- Semester examination and case study presentations

Continuous Assessment	Final Assessment		
25 - 50%	50 - 75%		
Details: quizzes %, Case studies%	Theory (%)	Practical (%)	Other (%)(specify)
0 - 10 %, 0 – 40 %	100%	0%	

Recommended Reading:

1. Soman, S, Swernofsky, N R, (1993), Experience Technology, Macmillan/McGraw-Hill.
2. Harms, HR, Kroon, D K, (1992), Production Systems Technology, Macmillan/McGraw-Hill.
3. Annual Reports, Central Bank of Sri Lanka.
4. Materials provided in the CAL.

Year of study 2:

Year of Study	2		
Semester	1		
Course Code	MGTE 21243		
Course Name	Marketing Management		
Pre-requisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	45	15	90

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Explain the increasingly significant role of marketing in modern organizational management
2. Identify the variables at work of various marketing situations
3. Apply c marketing theories to the demands of business and management practice.
4. Find and generate information/data needed to inform problem solving in marketing using appropriate methodology.
5. Prepare a comprehensive marketing plan for a business organization

Course Content:
Introduction to Marketing - Marketing Philosophies & Marketing Orientation, Marketing Environment, Consumer Buying Behaviour, marketing strategies - segmentation, targeting, and positioning (STP) Strategies, Product and Brand management, Pricing strategies, Place Strategies & Logistics, Integrated Marketing Communications, Service Marketing Mix , Industrial Marketing, Digital and social media marketing concepts, Internet marketing, social media, direct marketing, and sales promotions, Ethics and social responsibilities, Digital marketing

Teaching /Learning Methods:
Lectures, supervised practical sessions and guided self-study assignments

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment 25 - 50%	Final Assessment 50 - 75%		
Details: Assignments- 0 - 50 %	Theory (%) 50-90%	Practical (%) 0%	Other (0-10%) (attendance)

Recommended Reading:

1. Kotler, P (2015), *Marketing Management,* 15th Edition, Prentice Hall.
2. Boyd and Walker (2015), *Marketing Management.*
3. Wells W D, Moriarty S and Burnett J (2011), 9th Edition, *Advertising & IMC; principles and practice,* Prentice Hall.
4. Materials provided in CAL.

Year of Study	2		
Semester	1		
Course Code	MGTE 21233		
Course Name	Operations Management		
Pre-requisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	45	15	90

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Define the role of operations in an organization
2. Demonstrate basic competencies in planning, organising, controlling and improving manufacturing and service operations
3. Apply techniques of productivity and quality management

Course Content:
Introduction to operations management, Historical background and development of production management systems, Product design and process selection, Forecasting techniques, Strategic capacity planning, Plant location, Facility layout, Aggregate planning, Operations scheduling, Inventory control, Material Requirements Planning, Product quality, Quality management, Quality Control charts, acceptance sampling, Work Study, learning effects, lean production, Recent trends in operations management

Teaching /Learning Methods:
Lectures, case analysis and industry visits, Product Design and Development Competition

Assessment Criteria:
End-of-semester examination and continuous evaluation, Product Design and Development Competition

Continuous Assessment 40%	Final Assessment 60%	
Details: Group Project 10%, Product Design and Development Competition 30%	Exam : 90%	Attendance: 10%

Recommended Reading:

1. Stevenson W J, (2018), Operations Management, 13th Edition, McGraw Hill.
2. Chase, J and Aquilano, (2006), Operations Management for competitive advantage,11th Edition, McGraw Hill –Irwin.
3. Slack, N, Johns, A.B., Johnston, R., Operations Management, Seventh Edition, Pearson.
4. Krajewski, L J, Ritzman, L P & Malhotra M K, (2016), Operations management: processes and supply chains, 11th edn/global edn, Pearson Education, Inc..
5. Material provided in CAL.

Year of Study	2		
Semester	2		
Course Code	MGTE 22273		
Course Name	Human Resource Management & Leadership Communication		
Pre-requisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown	Theory	Tutorial/ Practical	Guided Self-study
(Notional Hrs.)	45	30	75

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Describe the importance of Human Resource Management and the changing role of Human Resource Professionals.
2. Demonstrate an understanding of the importance of Human Resource Management activities and their implications on the organization's performance.
3. Discuss the emerging concepts of employee talent management and knowledge management.
4. Define leadership and identify the qualities and characteristics of effective leaders.
5. Analyze transformations in personality, attitudes, and skills.
6. Acquire effective presentations, report writing and listening skills.
7. Demonstrate communicational skills in daily and professional lives.

Course Content:
Introduction to Human Resource Management, Job Analysis, Human Resource Planning, Recruitment and Selection, Selection Interview, Employee Performance Management, Career Management, Employee Rewards and Compensation, Employee Benefits, Employee Discipline and Grievances, Emerging Concepts in HRM, International Human Resource Management, Evolution of leadership thoughts, The traits theory, Contingency and situational theories, Path-goal behaviour of leaders, Pygmalion effect; Passive, aggressive, and assertive personalities, Childhood experience and formation of beliefs and attitudes, Influence of groups and environment on personality, charismatic and transformational leadership, Physical, clinical, psychiatric, and do- it-yourself methods of leadership development, Universal communication model, purpose and modes of organizational communications, and techniques of presentation, negotiation, writing and listening.

Teaching /Learning Methods:
Lectures, in-class discussions, real world case studies and self-study

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment 25 – 50%	Final Assessment 50 - 75%		
Details: quizzes %, mid-term %, Assignments% 0 - 30 %, 0 – 30 % 0 – 50 %	Theory (%) 10%- 90%	Practical (%) 0%	Other (%)(Attendance) 0%- 10%

Recommended Reading:

1. Armstrong M. and Taylor S., (2017), Armstrong's Handbook of Human Resource Management Practice: Building Sustainable Organisational Performance Improvement, 14E, Kogan Page.
2. Snell, S. Morris and Bohlander, G. W., (2015), Managing Human Resources, 17E, Cengage Learning.
3. Dessler, G., (2016), Human Resource Management, 15E, Pearson.
4. D. Torrington, L. Hall, C. Arkinson and S. Taylor, (2017), Human Resource Management, 10E, Pearson.
5. Mathis, R L and Jackson, J H, (2010), Human Resource Management, South-Western College Publishing, Florence.
6. Noel, Hollenbeck and Gerhart, (2005), Human Resource Management, McGraw-Hill, Noida.
7. Materials provided in CAL.

Year of Study	2		
Semester	2		
Course Code	MGTE 22263		
Course Name	Logistics and Supply Chain Management		
Pre-requisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	45	15	90

Course Aim/Intended Learning Outcomes:

On completion of this course, the student should be able to:

1. Discuss the evolution and fundamental concepts in supply chain and logistics with related theory and practices.
2. Identify key components in logistics.
3. Discuss the significance of all partners involved in a supply chain.
4. Develop a product supply chain map.
5. Discuss the significance of supply chain and logistics systems to other disciplines.
6. Identify key issues related to logistics and supply chain.

Course Content:

Introduction to logistics, Evolution of logistics, Supply chain management, Mapping a product supply chain, Logistics functions, 7Rs in logistics, Supply chain partners, Supply chain flows, Upstream and downstream supply chain, Inbound and outbound logistics, Horizontal and Vertical Integration in Supply Chain, Contribution of the logistics sector to the national economy, Development of the logistics sector in Sri Lanka and key players, Key activities in logistics, Warehousing, Transportation, Inventory management, Demand planning, Distribution planning, Global supply chain, World commodity markets, Global Supply Chains; Supply Chain Strategies (Lean, Agile, Leagile, Resilience, Digital, Sustainable, Survivable, Viable); Supply Chain Planning, Demand and Order Management (Basic understanding of Inventory Models, EOQ calculations), Key issues related to logistics and supply chain

Teaching /Learning Methods:

Lectures, home-work, and case study analyses

Assessment Criteria:

End of course unit examination and continuous assessments

Continuous Assessment	Final Assessment		
0 – 50%	50 - 75%		
Details: In-class activities%, Case studies %	Theory (%)	Practical (%)	Other (%)
0 - 10 %, 0 – 40 %	10%- 90%	0%	

Recommended Reading:

1. Rahul, V A, (2005), Supply Chain Management and Cases, PHI Learning, New Delhi.
2. John, IC and Robert, A N, (2009), A Logistics Approach to Supply Chain Management, Cengage Learning.
3. Coyle, J.J., Langley, C.J., Gibson, B.J., Novack,R.A., Bardi, E.J. 2017. Supply Chain Management - A Logistics Perspective, 10e, South-Western, Cengage Learning, Australia.
4. Joel, D W, Keong, L and Keah Choon, T, (2012), Principle of Supply Chain Management – A Balanced Approach, Cengage Learning.
5. Ronald, H B, Samir, K S, (2007), Business Logistics/ Supply Chain Management, Pearson Education.
6. David, JB, Stephen, L and Joe, B H, (2002), Logistics, PHI Learning, New Delhi.
7. Paul, R M and Donald, F W, (2008), Contemporary Logistics, PH of India.
8. Material provided in CAL.

Year of study 3:

Year of Study	3		
Semester	1		
Course Code	MGTE 31373		
Course Name	Project Management		
Pre-requisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	40	15	105

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Assess the critical role played by key project management processes for the successful completion of a project
2. Explain the phases of the project management lifecycle
3. Develop a project plan covering scope, schedule, budget, risks and resource allocation
4. Determine and interpret various project performance metrics
5. Explain the impact of new practices, such as agile practices, on managing projects
6. Demonstrate the use of MS Project for managing projects

Course Content:
The project manager's role, Internal and external environment, Project structures, Project and product life cycles, Agile software development, PM processes, Rationale and business case, Portfolio management, Financial analysis, Budgeting agile projects, The Charter, The Triple constraints, Project constraints and assumptions, Statement of Work (SOW), Scoping agile projects, WBS, Cost estimation, Stakeholder management and project communication, The network and the critical path, Project risk management, project performance management, Earned Value Management, project quality, procurement management, Ethics and professional responsibility

Teaching /Learning Methods:
Lectures, supervised hands-on practical sessions, and group work

Assessment Criteria:
End-of-semester examination and continuous assessment

Continuous Assessment 40 – 60%	Semester-End Examination 60 – 40%		
Details: Assignments: 50% MCQs: 50%	Theory (%) 100%	Practical (%)	Other (%)(specify)

Recommended Reading:

1. Jack T. Marchewka, 2016, Information Technology Project Management: Providing Measurable Organizational Value, 5th Edition, John Wiley
2. Clifford F. Gray and Erik W. Larson, (2020), Project Management: The Managerial Process, 8th Edition, McGraw-Hill
3. Kathy Schwalbe, (2017), An Introduction to Project Management, Sixth Edition, CreateSpace Independent Publishing Platform
4. Material provided in CAL

Year of Study	3		
Semester	1		
Course Code	MGTE 31383		
Course Name	Research Methods		
Pre-requisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	30	45	75

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Evaluate the importance of conducting a research for the identified problem.
2. Assess suitable research design for a given research problem.
3. Demonstrate a grasp of the conceptual framework necessary to formulate and conduct a research.
4. Prepare a detailed research proposal.
5. Perform research design, data gathering, analysis and presentation of conclusions and recommendations.
6. Analyse data using Partial Least Squares Structural Equation Modeling (PLS-SEM).

Course Content:
Some reflections on the theory of evolution of knowledge, Inductive and deductive methods in research, Research design**:** identifying issues and problems, defining research problem**(s)** and objectives, literature review, Identifying data requirements, sources, and instruments for data gathering.
An introduction to Structural Equation Modelling (SEM), Measurement Model vs Structural Model of a path model, Reflective vs Formative measurement models, Mediation effect and Moderation effect analysis, and Hierarchical Component Model analysis.

Teaching /Learning Methods:
Lectures, interactive classroom sessions and case analysis

Assessment Criteria:
End of course unit examination and case study presentations

Continuous Assessment 40 - 60%	Final Assessment 40 - 60%		
Details: Assignments% 40 – 60 %	Theory (%) 100%	Practical (%)	Other (10%)(attendance)

Recommended Reading:

1. Blaxter, L, Hughes, C and Tight, M, (2010) How to Research 4th edition, McGraw-Hill Education.
2. Kumar R, (2011), Research Methodology – A step-by-step guide for beginners 3rd edition, SAGE Publications.
3. Sharp, J A, Peters, J and Howard, K, (2002), The Management of a Student Research Project, 3rd edition Gower Publishing Company.
4. Sekaran, U and Bougie, R, (2010), Research Methods for Business – A Skill Building Approach, Wiley.
5. Material provided in CAL.

Year of Study	3		
Semester	1		
Course Code	MGTE 31403		
Course Name	Management of Technology		
Pre-requisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Independent Study / Tutorial	Guided Self-study
	45	30	75

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Evaluate technology requirements of an organization.
2. Explain the need of technology management.
3. Describe the basic elements of technology and technology capabilities.
4. Explain the pattern of technology performance against time.
5. Explain the need and the mechanisms of technology planning and forecast.
6. Explain the rational and mechanisms of technology acquisition and technology transfer.
7. Discuss the role of research and development in technology development.
8. Explain how technology strategy relates to firm's long term objectives.

Course Content:
Characteristics of technology for socio-economic development, Components of technology and Technological capabilities, Technology life cycles, Technology forecasting/foresight and Technology Roadmapping process, Technology acquisition and Technology transfer, Innovation ecosystem and research and development (R&D) as a core of the technological innovation process, Business Strategy and Strategic management of technology, Technology competitiveness and technology planning, Introduction to Industry 4.0, Technology Entrepreneurship, Open Innovation, Reasons for selling/buying technology, Technology foresight case evaluation,

Teaching /Learning Methods:
Lectures and case discussions

Assessment Criteria:
End of course unit examination and continuous assessments.

Continuous Assessment	Final Assessment		
25 - 50%	50 - 75%		
Details: quizzes %, mid-term %, Assignments% 0 - 50 %, 0 – 50 % 0 – 50 %	Theory (%) 100%	Practical (%) 0%	Other (0-10%) (attendance)

Recommended Reading:

1. Burgelman, R A and Maidique, M A and Wheelright, C, (2001), Strategic Management of Technology and Innovation, Irwin, Chicago.
2. Khalil, T, (2000), Management of Technology: The Key to Competitiveness and Wealth Creation, McGraw-Hill Book Co, Singapore.
3. Sherif, H, Khalil, T M, (2006), Management of Technology: New Directions in Technology Management, Elsevier, Oxford.
4. Morel_Guimaraes L, Khalil, T, Hosni Y A, (2005), Management of Technology: Key Success Factors for Innovation and Sustainable Development, Elsevier, Oxford.
5. Hosni, Y A, Khalil, T M, (2004), Management of Technology: Internet Economy: Opportunities and Challenges for Developed and Developing Regions of the World, Elsevier, Oxford.
6. Material provided in CAL.

Year of Study	3		
Semester	1		
Course Code	MGTE 31413		
Course Name	Warehouse Management and Industrial Shipping		
Pre-Requisites	None		
Co-Requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Tutorial/ Practical	Guided Self-study
	45	15	90

Course Aim/Intended Learning Outcomes:

On completion of this course, the student should be able to:

1. Analyze the basic process of a warehouse operation
2. Analyze the KPIs related to warehouse operation
3. Evaluate warehouse operation models
4. Evaluate shipping options for industrial cargo
5. Analyze shipping markets and key indexes
6. Evaluate main operations related to industrial shipping process

Course Content:

Warehouse Process Management: Receiving, Put away, Storage, Order picking, Shipping, Distribution Management, Network Design, Warehousing models, Warehouse activity profiling, Measuring and benchmarking warehouse operations, Warehouse process pallet storage and retrieval systems, Store functions and responsibilities, Store procedures and planning, Store System costs and rates, Materials - Handling considerations and systems, Computerizing warehouse operations, Warehouse layout planning, Warehouse safety and security, Best practices, Audits and compliance, Warehouse location selection, Inventory management

Shipping market and players, Liner shipping industry, Tramp shipping industry, Freight market, Baltic Freight Index (BFI), Vessel new building market, Demolition market, Sale and purchase market, Liner agency services, Strategic alliances, consortia, Merger and acquisitions, Shipping networks, Seaport operations, Route planning, Fleet management, Port choice, Network design: Pendulum services, Round the world services, Ship owners, Insurance, Charter parties, Shippers and consignees, Freight forwarders, Chartering types, Voyage, time and bareboat chartering, Yard planning, Crew scheduling, Ship planning, Berth planning, Strategic, Port operating models, Tactical and operational level issues

Teaching /Learning Methods:

Lectures, interactive classroom sessions and case analysis

Assessment Criteria:

End of course unit examination and continuous assessments

Continuous Assessment 40%	Final Assessment 60%		
Details: Assignment%, Project 0 - 20 %, 0 – 20%	Theory (%) 10%- 90%	Practical (%) 0%	Other (%)(Attendance) 0%- 10%

Recommended Reading:

1. Edward, H F, (2004), World-Class Warehousing and material handling, TATA McGraw-Hill, New Delhi.
2. Rahul, V A, (2005), Supply Chain Management: Concepts and Cases, PHI Learning, New Delhi.
3. John, J C and Robert, A N, (2009), A Logistics Approach to Supply Chain Management, Cengage Learning.
4. Joel, D W, Keong, L and KeahChoon, T, (2012), Principle of Supply Chain Management – A Balanced Approach, Cengage Learning.
5. Ronald, H B and Samir, K S, (2007), Business Logistics/ Supply Chain Management, Pearson Education.
6. Paul R M, Donald F W, (2008), Contemporary Logistics, PH of India.
7. Tompkins, JA, White, Y A, Bozer, EH and Frazer, (1996), Facilities Planning, Wiley.
8. Peter Brodie, P. (2018) Commercial Shipping Handbook.
9. Lun, Y.H.V., Lai. K-H. and Cheng. T.C.E. (2010) Shipping and Logistics Management. Springer Link.
10. Levinson, M. (2008) The Box: How the Shipping Container Made the World Smaller and the World Economy Bigger.
11. Material provided in CAL.

Year of Study	3		
Semester	1		
Course Code	MGTE 31433		
Course Name	Computer Based Tools for Management Applications		
Pre-requisites	MGTE 12222 - Optimization Methods in Management Science		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	30	60	60

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Apply spreadsheet modelling using Visual Basic for Application (VBA) to model and solve Operations Research, Operations Management, Statistics, Supply chain Management, Economics, Human resource Management, and Finance real world complex problems.
2. Improve student competence in utilizing Excel as a decision-making support tool, emphasizing the use of Excel to document and demonstrate models.
3. Use Microsoft PowerBI and R tools as data driven decision making tools to visualize and solve practical problems in industry.
4. Identify opportunities and apply Microsoft Power Apps as workflow automation and management tools to smoothen the ongoing activities of organizations.

Course Content:
Introduction to Spreadsheet Modelling using VBA, Introduction to Optimization Modelling using VBA, Spreadsheet Engineering and Managerial decision modelling, Decision Making under Uncertainty, VBA applications for simulation, Goal Seek, Data Tables/Scenario Manager, Solver, Applications of spreadsheet-based decision support systems, Excel Data Analysis, Introduction and Applications of Microsoft PowerBI, Integration of R programming into Microsoft PowerBI, Microsoft Power Apps.

Teaching /Learning Methods:
Lectures, in-class discussion of cases, self-study and mini-projects

Assessment Criteria:
End-of-course exam, in-class assignments, mini-project, Quizzes

Continuous Assessment 60%	Final Assessment 40%	
Details: Group Project 30%, Mini Project 30%	Exam : 90%	Attendance: 10%

Recommended Reading:

1. Winston, W and Christian, A S, (2009), Practical Management Science, Revised, 3rd Edition.
2. Winston, W L, (2007), Microsoft® Office Excel® 2007: Data Analysis and Business Modelling, Microsoft Press.
3. Christian A S, (2007), VBA for Modelers: Developing Decision Support Systems with Microsoft® Excel, Thomson, Brooks/Cole.
4. Stephen, G P and Kenneth, R B, (2010), Management Science The Art of Modeling with Spreadsheets3rd Edition.
5. Weston, M and Martín, E B, (2023), Learn Microsoft Power Apps: The definitive handbook for building solutions with Power Apps to solve your business needs, 2nd Edition.
6. Material provided in CAL.

Year of Study	3		
Semester	1		
Course Code	MGTE 31443		
Course Name	Strategic Marketing and International Trade		
Pre-requisites	MGTE 21243 Marketing Management		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	45	Case study discussions within lecture hours	105

Course Aim/Intended Learning Outcomes:

After completing this module, the students should be able to,

1. Compare and contrast the key principles of marketing strategy.
2. Explain marketing and strategy concepts and ideas in their own words.
3. Think strategically about marketing issues and provide recommendations.
4. Differentiate between domestic and international trade.
5. Evaluate international markets in relation to a given product(s), and Advise SMEs to develop strategies for exporting.
6. Conduct marketing research, and Develop a marketing plan for a company.
7. Deliver an oral presentation in a professional, engaging manner.

Course Content:

Innovative and competitive marketing strategies-Marketing strategies for the 21st Century- Marketing Thought and Practice, and Societal Behaviour; Strategic Analysis – Business, Industry & Market need and Analysis , Review of Internal Capabilities, Problems and Opportunities; Strategic decisions at: corporate, SBU and functional level; Sensory branding – Major elements of branding (5 sense) and how to compete with international brands, Ccompetitive pricing strategies for export; Need creation for innovative products - competitive strategies and product development strategies, Definitions and the concept of global marketing, Economic, political, legal, socio-cultural environment; International Marketing Management – Global marketing strategy, International Marketing Intelligence; International product policy, Distribution strategy, International promotion, International trade requirements (Pricing, Certification requirement etc.); The future of International Marketing and Sri Lankan Export activities – Export policy and development, Export development institutions, Operational aspects of International Trade, TARIFF and Non-TARIFF barriers

Teaching /Learning Methods:

Lectures, supervised practical sessions and guided self-study assignments.

Assessment Criteria:

End of course unit examination and continuous assessments.

Continuous Assessment 25 - 50%	Final Assessment 50 - 75%		
Details: Assignments- 0 - 50 %,	Theory (%) 90%	Practical (%) 0%	Other (0-10%) (attendance)

Recommended Reading:

1. Kotler, P, (2006), Management, 12thEdition, Prentice Hill, Delhi.
2. Boyd and Walker, (2007), Marketing Management, 3rd Edition, McGraw Hill.
3. Onkvisit, S and Show, J J, (1996), International Marketing, Analysis and Strategy, 3rd Edition, Prentice Hall.
4. Material provided in CAL.

Year of Study	3		
Semester	1		
Course Code	MGTE 31423		
Course Name	Advanced Optimization Methods in Management Science		
Pre-requisites	MGTE 12263 - Optimization Methods in Management Science		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	15	45	40

Course Aim/Intended Learning Outcomes:

On completion of this course, the student should be able to:

1. Analyse, develop and use algorithms to tackle optimization problems of diverse kinds
2. Apply appropriate probabilistic and other quantitative techniques and methodologies to transform managerial situations including those under uncertainty, into optimization models.
3. Use the concepts discussed to analyze different types of business problems.
4. Analyse available optimization approaches and use the software to solve complicated problems.

Course Content:

Stochastic Markov process: Transition matrices and steady state vector, absorbing Markov process

Dynamic Programming: Introduction to Dynamic Programming under certainty and under uncertainty, Infinite State Dynamic Programming,

Multicriteria decisions: Goal Programming and Multiple Objective Optimization

Defining the Decision Variables, Defining the Goals, Defining the Goal

Constraints, GP Objective Functions, Implementing the Model, Implementing the revised model, Solving complex models,

Game Theory: two-person zero-sum games with and without saddle points, mixed strategy games, m*n strategies LP solution of zero-sum games

Queuing Theory/Waiting Line Theory: Waiting line situations in practical life, arrival distribution, service distribution, queue discipline, single channel and multiple channels waiting line model with Poisson arrivals, single channel and multiple channel waiting line model with Poisson arrivals and arbitrary service times. Limited waiting room size, limited population, Economic analysis of waiting line.

Teaching /Learning Methods:

Lectures, interactive classroom sessions, case discussions, and applications of Microsoft Excel Solver

Assessment Criteria:

End of course unit examination, group assignments, mid-term examination, practical test, presentations class attendance.

Continuous Assessment	Final Assessment		
40% - 50%	50% - 60%		
Details: quizzes 0-20%, mid-term test 0-20 %, practical test 0-20%	Theory (%) 60-90%	Practical (%) 10-30%	Other (attendance) 10%

Recommended Reading:

1. Anderson, D R, Williams, T A and Sweeney, S J, (2011), An Introduction to Management Science: Quantitative Approaches to Decision Making, 13th Edition, South Western College Pub.
2. Fedrick, S. Hiller, G. Lieberman, B. Nag, (2014), Introduction to operations research, McGraw-Hill Professional, 9th Edition, McGraw Hill Education (India) Pvt. Ltd.
3. Kenneth R. Baker, (2015), Optimization Modeling with Spread sheets, Thompson Learning Academic Resource Center
4. Barry, R and Ralph, M S, (2015), Quantitative analysis for Management 12th Edition Prentice Hall.
5. Materials provided in eKel.

Year of Study	3		
Semester	1		
Course Code	MGTE 31293		
Course Name	Computer Integrated Manufacturing		
Pre-requisites	MGTE 12273 - Industry and Technology MGTE 21233 - Operations Management		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Independent Study / Practical	Guided Self-study
	45	70	35

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Demonstrate understanding of the scope for computer integration in the design and operation of manufacturing systems.
2. Demonstrate knowledge of computer-based systems for product design and process planning in manufacturing.
3. Demonstrate knowledge of computer-based systems employed in manufacturing.
4. Develop and apply computer-aided techniques for manufacturing planning and control.
5. Identify and manage the impact of automation in manufacturing systems.

Course Content:
Scope for Computer Integration in manufacturing (CIM), elements of CIM systems
Computer based product and process planning concepts, Computer Aided Draughting and Design (CAD) software;
Production Flow Analysis algorithms, Group Technology, Computer aided layout planning techniques and software;
Programmable machinery, Principles of Computer Numerically Controlled (CNC) machines and programming, Elements of Computer Aided Manufacturing (CAM) systems; vision systems
Automatic control systems and instrumentation, Automated measurement and inspection;
Algorithms for computer aided production planning and control;
Issues of managing advanced manufacturing technologies

Teaching /Learning Methods:
Lectures, interactive classroom sessions and case analysis.

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment 25 - 50%	Final Assessment 50 - 75%		
Details: quizzes %, Assignments% 0 - 50 %, 0 – 50 %	Theory (%) 100%	Practical (%) 0%	Other (0-10%) (attendance)

Recommended Reading:

1. Rehg J S and Kraebber H W, (2000), Computer Integrated Manufacturing, Prentice Hall International.
2. Tompkins, J A, White, Y A, Bozer, E H and Frazer, (1996), Facilities Planning, Wiley.
3. Jean, B W, (1992), CIM Principles of Computer Integrated manufacturing, Wiley.
4. David, D B, Mark, R H and Philip W, (1991), Computer-integrated design and manufacturing, McGraw-Hill
5. Material provided in CAL.

Year of Study	3		
Semester	1		
Course Code	MGTE 31303		
Course Name	Procurement and Supply Management		
Pre-requisites	MGTE 21233 - Operations Management MGTE 22263 - Logistics and Supply Chain Management		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Tutorial/ Practical	Guided Self-study
	45	15	90

Course Aim/Intended Learning Outcomes:

On completion of this course, the student should be able to:

1. Demonstrate understanding of the process and role of procurement management
2. Explain the fundamentals of supply management and supply chain management
3. Evaluate procurement-related functional strategies for effectiveness
4. Analyze and evaluate the suppliers and performance
5. Explain the approaches for managing supplier relationship
6. Understand the role of inventory, delivery, and logistics in purchasing
7. Evaluate the risk involved in purchasing
8. Understand the emerging trends in procurement and supply management

Course Content:

Introduction to purchasing, procurement, supply management, supply chain management, Value chain, organization structure for purchasing, purchasing strategy, purchasing process, Contrasting Approach to Supply, make or buy decision, insourcing, outsourcing, Just-in-time systems, Vendor managed inventory, Postponement, Pull, Push, Push-pull strategies, Global sourcing, Supply chain planning, Bullwhip effects, 3PL, supplier selection and evaluation, planning, specification, value analysis, managing supplier relationship, negotiation, inventory management, delivery and logistics, reverse logistics, procurement risk, contract management, purchasing policy, sustainability, E-procurement and emerging trends, e-SCMs (MRP, SRM, CRM, DRP, ERP, POS, EDI, RFID, Bid optimization), Supply Chain Performance Management (SCOR model 12.0-Linking to digital supply chain), INCO Terms, Overview for the Legal aspects of international trade).

Teaching /Learning Methods:

Lectures, Quizzes, Home-work, and case study analyses

Assessment Criteria:

End of course unit examination and continuous assessments

Continuous Assessment	Final Assessment		
0 – 35%	50 - 75%		
Details: Research Publication %, In-class activities%, Group presentations% 0 - 15 %, 0 – 7.5%, 0 – 7.5%	Theory (%) 10%- 90%	Practical (%) 0%	Other (%) (Attendance) 0%- 10%

Recommended Reading:

1. Johnson, P F, Leenders, M R and Flynn, A E, (2011), Purchasing and Supply Management, McGraw Hill, New York.
2. Lysons, K & Farrington, B. 2006, Purchasing & Supply Chain Management, 8th edition, Pearson publication.
3. Monczka, R.M., Handfield, R.B., Giunipero, L.C. and Patterson, J.L. (2009) Purchasing and Supply Chain Management, Fourth Edition, South Western
4. Monczka, Trent and Hardfield, (2002), Purchasing and Supply Chain Management, Thomson, South Western.
5. Victor, H P and David, J P, (1997), Purchasing and Supply Chain Management: Creating Vision Thomson Publishing, New York.
6. Material provided in CAL.

Year of Study	3		
Semester	1		
Course Code	MGTE 31393		
Course Name	Managerial Finance		
Pre-requisites	MGTE 12253 - Accounting Concepts and Costing		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Independent Study / Practical	Guided Self-study
	45	45	60

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Describe the differences between the different branches of accounting and financial management
2. Explain Principal Agent problem and the managerial implications
3. Develop the cash flow statement
4. Use financial ratios to analyze the performance of a company
5. Use present value / future value concepts to understand and design financial products
6. Value long terms financial products, bonds and stocks
7. Evaluate capital investments using capital budgeting techniques.

Course Content:
Overview of Financial Management, Financial Statements, Taxes and Cash Flows, Working with Financial Statements, Principles of Valuation, Valuing Long Term Securities, NPV and Investment Criteria, Capital Budgeting, Risk & Return

Teaching /Learning Methods:
Lectures, classroom discussions, group study and guided self-study assignments.

Assessment Criteria:
End of course unit examination and continuous assessments.

Continuous Assessment	Final Assessment		
20 - 40%	60 - 80%		
Details: Quizzes %, mid-term %, Assignments% 0 - 40%, 0 – 40% 0 – 40%	Theory (%) 100%	Practical (%) 0%	Other (0-10%) (attendance)

Recommended Reading:

1. Ross S A, Westerfield R W, Jordan B D, (2021), Fundamentals of Corporate Finance, 13th Edition, McGraw-Hill Publishing Company Limited.
2. Brealey, R A, Myers, S C, Allen, F, and Adams, A, (2023), Principles of Corporate Finance, 14th Edition, McGraw-Hill.
3. Ronald Hilton, (2022), Managerial Accounting: Creating Value in a Dynamic Business Environment, McGraw-Hill Publishing Company Limited.
4. Daily Financial Times, seen at www.ft.lk
5. Material provided in CAL.

Year of study 4:

Year of Study	4		
Semester	1		
Course Code	MGTE 41373		
Course Name	Strategic Management		
Pre-requisites	MGTE 11233 Business Statistics and Economics, MGTE 21243 Marketing Management, MGTE 21233 Operations Management, MGTE 31393 Managerial Finance		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Independent Study / Practical	Guided Self-study
	45	30	75

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Explain the need and process of strategic planning in an enterprise.
2. Conduct and draw conclusions from external analysis of an organization's environment.
3. Conduct and draw conclusions from internal analysis of an organization's capabilities.
4. Formulate strategies to exploit market opportunity.
5. Develop implementation plans to execute the strategy.
6. Draw up a business plan.

Course Content:
Understanding the need for strategic planning; Process of strategic planning; Components of strategic planning; Developing vision and mission statements; Analysis of the external and internal analysis; Competitive analysis; Driving forces and critical success factors; Formulation of objectives; Different strategy options and selection.

Teaching /Learning Methods:
Lectures, in-class assignments / presentations, doing research and formulating a strategic plan for a business and pitching to potential investors.

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment 50 - 70%	Final Assessment 30 - 50%		
Details: Quizzes %, Assignments% 0 – 50% 0 – 50%	Theory (%) 100%	Practical (%) 0%	Other (0-10%) (attendance)

Recommended Reading:

1. Hill, C W L, Schilling, M A, Jones, G R, (2023), Strategic Management Theory: An Integrated Approach, Theory and Cases, 14th Edition, Boston MA Cengage.
2. Rothaermel, Frank T, (2023), Strategic Management, 6th Edition, McGraw-Hill.
3. Pearce, J A and Robinson, R B, (2014), Strategic Management – Planning for Domestic and Global Competition, 14th Edition, McGraw-Hill/Irwin.
4. Daily Financial Times – www.ft.lk
5. Material provided in CAL.

Year of Study	4		
Semester	1		
Course Code	MGTE 41323		
Course Name	Professional Practices		
Pre-requisites	MGTE 11243 - Principles of Management & Organizational Behaviour		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	45	30	75

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Describe the characteristics of a professional
2. Evaluate on the nature of professionalism and ethical challenges inherent in professionalism
3. Discuss and critique prominent ethical frameworks
4. Analyze ethical challenges and dilemmas through ethical frameworks
5. Identify reasons why people may be unethical
6. Relate ethical concepts and materials to ethical problems in industry / society
7. Use a framework to assess need for whistle blowing

Course Content:
Professionalism and Professional conduct; Ethics, Differentiation between law, morals and ethics, professionalism and code of ethics, ethics and professional judgment, professional judgment and social intelligence, Whistleblowing; Introduction to decision making, privacy, intellectual property

Teaching /Learning Methods:
Lectures, practical assignments, case analysis, group study and guided self-study assignments

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment	Final Assessment		
25 - 60%	40 - 75%		
Details: quizzes %, mid-term %, Assignments%	Theory (%)	Practical (%)	Other (%)(specify)
0 - 60%, 0 – 60% 0 – 60%	100%	0%	

Recommended Reading:

1. Joseph DesJardins, (2020), An Introduction to Business Ethics, 6th Edition, McGraw Hill.
2. David Kelly, Debbie Hutchins, (2020), The Art of Reasoning – An Introduction to Logic and Critical Thinking, 5th Edition, W W Norton and Company.
3. Bazerman, M H and Moore, D A, (2009), Judgment in Managerial Decision Making, 7th Edition, John Wiley.
4. Ethics for the Professions, (2003), John Rowan & Samuel Zinaich, Jnr. Wadsworth.
5. Material provided in CAL.

Year of Study	4		
Semester	1		
Course Code	MGTE 41333		
Course Name	Business Process Engineering		
Pre-requisites	MGTE 21233 - Operations Management		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown	Theory	Practical	Guided Self-study
(Notional Hrs.)	45	30	75

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Demonstrate understanding of the link between business strategy and the business process.
2. Identify the need for creating business processes using industry best practices for sustaining competitive advantage.
3. Model and analyse business processes using formal business modelling languages.
4. Identify improvement needs and conduct BPR exercise.
5. Apply methods of benchmarking.
6. Apply models for performance measurement.

Course Content:
Business strategy, Visionary leadership in business, Organisation for full potential, , facing competition, world-class organisations and practices, best-practice reference models,
Business strategy and competitiveness, Need for business re-engineering (BPR), BPR roadmap, Business modelling methods, Tools for business process analysis and improvement, capability Maturity Model (CMM), Types of Benchmarking, Selection of partners, Performance Measurement Models, Balanced Score Card (BSC), Business Analysis Tools (Business Platform, So Is My Boss)

Teaching /Learning Methods:
Lectures, interactive classroom sessions and case analysis

Assessment Criteria:
End-of- Semester examination and case study presentations

Continuous Assessment 40%		Final Assessment 60%	
Details: Quizzes %, mid-term %, Assignments% 0 - 40%, 0 – 40% 0 – 40%	Theory (%) 0-100%	Practical (%) 0-20%	Other (%)(specify)

Recommended Reading:

1. Handbook of Industrial and Systems Engineering, Salvendy et.al.
2. Chase, A and Jacobs, (1998),Operations Management: Manufacturing and Services,8th Edition, McGraw Hill – Erwin.
3. Slack, N, Chambers, S, and Johnston, R, (2000),Operations Management, 3rd Edition, Prentice Hall.
4. Stevenson, W J, (2001),Operations Management, 7th Edition, McGraw Hill – Irwin.
5. Carnegie Mellon Univ, Software Engineering Inst,, Mark, C P, Charles, V W, Bill, C, Mary, B C (1994), The Capability maturity Model, Guidelines for Improving Software process, Person Education.
6. Mitra, A, (1998), Quality Control and Improvement, 2nd Ed, Pearson Education, Asia.
7. Handbook on Business Process Management 1: Introduction, Methods, and Information Systems (International Handbooks on Information Systems).
8. Material provided in CAL.

Year of Study	4		
Semester	1		
Course Code	MGTE 41313		
Course Name	Statistical Data Modelling		
Pre-requisites	MGTE 11233 – Business Statistics and Economics		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical/ Tutorial	Self-Guided Study
	15	45	40

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. provide an understanding of statistical models for handling common data analysis problems;
2. develop skills for fitting, interpreting and assessing statistical models;
3. perform exploratory data analysis with descriptive statistics on given datasets;
4. produce models for predictive statistical analysis;
5. develop computer skills for exploring and modelling different kinds of data;
6. interpret results for a variety of models using spreadsheets and the R programming language

Course Content:

Sampling techniques, Data preprocessing, Exploratory Data Analysis: Data visualization techniques, descriptive statistics, identifying data patterns

Statistical inference: Population parameters and statistics, One-sample tests of hypothesis and confidence interval Two-sample tests of hypothesis

Data Analysis and Modelling: one-way ANOVA between groups, one-way repeated ANOVA, two-way ANOVA

Association and relationships: Correlation analysis, Simple linear regression and residual analysis, Chi-square test

Multivariate analysis: Hierarchical multiple regression analysis, Standard multiple regression analysis, Principal Component Analysis

Classification techniques

Teaching /Learning Methods:
Lectures, supervised practical sessions and guided self-study assignments

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment 40% - 50%	Final Exam 50% - 60%		
Details: quizzes 0-20%, mid-term test 0-20 %, practical test 10-20%	Theory (%) 70-90%	Practical (%) 10-20%	Other (attendance) 0 -10%

Recommended Reading:

1. Use of RStudio and various R packages are required. These are all freely available.
2. The textbook "Extending the Linear Model with R" by Faraway (2nd edition, 2016).
3. Agresti, Franklin and Klingenberg (2018, 4th edition). Statistics: the art and science of learning from data, Pearson.
4. G. James, D. Witten, T. Hastie and R. Tibshirani, ``An Introduction to Statistical Learning with Applications in R'' (http://www-bcf.usc.edu/~gareth/ISL).
5. Stefano Bonnini, Livio Corain, Marco Marozzi, Luigi Salmaso (2014) Nonparametric Hypothesis Testing: Rank and Permutation Methods with Applications in R
6. Materials provided in the eKel

Year of Study	4		
Semester	1		
Course Code	MGTE 41233		
Course Name	Corporate Finance		
Pre-requisites	MGTE 31393 – Managerial Finance		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Independent Study / Practical	Guided Self-study
	45	45	60

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Explain the working of financial markets
2. Analyze risk and return relationship, calculate portfolio risk and return
3. Describe features of long term financing, corporate securities, mechanics of a rights offer and calculate the value of a rights offering
4. Explain reasons for underpricing
5. Calculate cost of capital for a firm
6. Explain and analyze capital structure policy and implications for firm value
7. Analyze the impact of dividend policy on firms
8. Analyze the impact of working capital policy on the firm
9. Prepare a cash budget and use break even analysis for decision making
10. Analyze the buy versus lease decision
11. Apply and use financial options to manage firm risk

Course Content:
Behaviour of capital markets, Long term financing, Dividend policy, Portfolio risk and return, Capital structure and cost of capital, Cash Budgeting, Breakeven analysis, Working capital management, Leasing, Risk management and introduction to futures and options.

Teaching /Learning Methods:
Lectures, student led chapter discussions, presentations, guest lectures, practical assignments, case analysis, group study and guided self-study assignments

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment 25 - 50%	Final Assessment 50 - 75%		
Details: quizzes %, mid-term %, Assignments% 0 - 50%, 0 – 50% 0 – 50%	Theory (%) 100%	Practical (%) 0%	Other (%)(specify)

Recommended Reading:

1. Ross S A, Westerfield R W, Jordan B D, (2021), Fundamentals of Corporate Finance, 13th Edition, McGraw-Hill Publishing Company Limited.
2. Brealey, R A, Myers, S C, Allen, F, and Adams, A, (2023), Principles of Corporate Finance, 14th Edition, McGraw-Hill.
3. Ronald Hilton, (2022), Managerial Accounting: Creating Value in a Dynamic Business Environment, McGraw-Hill Publishing Company Limited.
4. Daily Financial Times, seen at www.ft.lk
5. Material provided in CAL.

Year of Study	4		
Semester	1		
Course Code	MGTE 41343		
Course Name	Logistics System Analysis and Geomatics		
Pre-requisites	All compulsory modules for OSCM		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown	Theory	Tutorial/ Practical	Guided Self-study
(Notional Hrs.)	45	15	90

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Identify the challenges faced by the logistics and transportation industry
2. Discuss theories and practices in managing complex transport systems
3. Collect and analyze logistics and transport-related data
4. Formulate and solve logistics and transport problems
5. Optimize the use of resources attached to the transportation industry
6. Discuss the significance of geography in the logistics and transportion context
7. Analyse geographic data for planning, operations, and development of logistics system
8. Incorporate geographical analysis in decision making related to logistics and transport
9. Apply choice theories for transport mode and routes

Course Content:
Descriptive Analytics, Diagonistic analysis, Predictive Analytics, Prescriptive Analytics, Spatial and temporal variations in transport demand, Carrier decision-making problems, Shipper decision-making problems, Fleet management, Career management, Routing and scheduling vehicle, Carrier selection, Mode choice, Unit load concepts, Shipment planning, Freight management, KPIs and Performance measurement, Intelligent Transport Systems, Long-haul freight transportation, Short-haul freight transportation, Deterministic and stochastic models, Driver assignment problem, Shipment consolidation and dispatching, Vehicle allocation problem, City Logistics, Multi-model transport
Transport geography, Impacts of geographical features on transport and logistics, Land-use models, Hinterland and foreland, Accessibility, Landlocked areas, Global logistics hub, Cross-border transportation, Transport connectivity, Transport networks, Urbanization, Spatial dynamics, GIS applications, Developing maps, Locational analysis, Network design, Gravity models, Logit models, Demand management, Forecasting, Mathematical modelling of logistics problem, Four steps transport model (Trip generation, attraction, mode choice, trip assignment), Network design, International shipping routes, Origin-destination matrices, Traffic flow theory, Activity-based modeling

Teaching /Learning Methods:
Lectures, interactive classroom sessions and case analysis

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment	Final Assessment		
0 – 40%	60-80%		
Details: Assignment%, Project	Theory (%)	Practical (%)	Other (%)(Attendance)
0 - 20 %, 0 – 20%	10%- 90%	0%	0%- 10%

Recommended Reading:

1. Ghiani, G., Laporte, G. and Musmanno, R. (2004) Introduction to Logistics Systems Planning and Control
2. V.A Profillidis and G.N. Botzoris (2018) Modeling of Transport Demand, Analyzing, Calculating, and Forecasting Transport Demand
3. Chopra, S and Peter, M (2013) Supply Chain Management, Strategy, Planning, And Operation. Pearson
4. John, IC and Robert, A N, (2009), A Logistics Approach to Supply Chain Management, Cengage Learning.
5. Material provided in the CAL

Year of Study	4		
Semester	1		
Course Code	MGTE 41303		
Course Name	Enterprise Systems		
Pre-requisites	INTE 21313 - Business Information Systems		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	30	45	75

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Describe the role of enterprise systems in carrying out processes in an organization.
2. Evaluate the costs and benefits of implementing an enterprise system.
3. Explain how "best practices" are incorporated in enterprise systems.
4. Explain the enterprise systems life cycle and discuss major decisions to be made at various points of time.
5. Evaluate the major enterprise system software providers and their packaged systems.
6. Examine the challenges associated with selecting and implementing enterprise system packages.
7. Examine current trends related to enterprise systems.

Course Content:
Business processes and business process integration, making the case for acquiring and implementing enterprise systems, analyzing business requirements for selecting and implementing an enterprise system, risk management strategy, selection of enterprise systems software, vendor selection, challenges associated with the implementation of enterprise systems applications, Integration issues, master data management, organizational change and change management, user commitment, communications, training. future issues and trends, working with development environment

Teaching /Learning Methods:
Lectures, supervised practical sessions and guided self-study assignments.

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment 25 - 50%	Final Assessment 50 - 75%		
Details: quizzes %, mid-term %, Assignments% 0 - 25 %, 0 – 25 % 25 – 50 %	Theory (%) 100%	Practical (%) 0%	Other (%)(specify)

Recommended Reading:

1. Ellen Monk and Bret Wagner, (2012), Concepts in Enterprise Resource Planning, 4th Edition, Cengage Learning.
2. Luvai Motiwalla and Jeffrey Thompson, (2014), Enterprise Systems for Management, 2nd Edition, Pearson Higher Education.
3. Marianne Bradford, (2106), Modern ERP: Select, Implement, and Use Today's Advanced Business Systems, 3rd edition, Lulu Press.
4. Enterprise Systems Architecture, Daljit Roy Banger, (2022), APress
5. Material provided in CAL.

Year of Study	4		
Semester	1		
Course Code	MGTE 41383		
Course Name	Advanced Operations Management		
Pre-requisites	MGTE 21233 - Operations Management		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown	Theory	Practical	Guided Self-study
(Notional Hrs.)	45	15	90

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:
1. Demonstrate understanding of the strategic role of operations management in an organization.
2. Apply concepts and techniques of operations management in manufacturing and services.
3. Analyse and evaluate the performance of operations function.
4. Model functions of operations systems using information technology.

Course Content:
Operations/ Manufacturing Strategy for competitiveness, Performance measurement and improvement, Performance Importance Matrix, Statistical Process Control, Acceptance sampling procedures, Manufacturing Resources Planning Systems (MRP II), Distribution Requirements Planning, Optimised Production Technology, Bull-whip effect
Production Control techniques, delivery performance, Modelling and simulation in operations management, issues of computerisation

Teaching /Learning Methods:
Lectures, assignments, case studies and self-study

Assessment Criteria:
End-of-course exam, mid-semester exam, in-class quizzes, mini group project

Continuous Assessment	Final Assessment
40%	60%
Details:	

Recommended Reading:

1. Chase, A and Jacobs, (2006), Operations Management – For Competitive Advantage, 11th edition, McGraw-Hill.
2. Slack, N., Brandon-Jones, A. and Johnston, R., (2010), Operations Management. 6th ed. Essex: Prentice - Hall.
3. Stevenson W J, (2018), Operations Management, 13th Edition, McGraw Hill.
4. Material provided in CAL.

Year of Study	4		
Semester	1		
Course Code	MGTE 41353		
Course Name	Sustainability and Economics in Logistics and Supply Chain		
Prerequisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Tutorial/ Practical	Guided Self-study
	45	30	75

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:

1. Recognize global sustainability issues associated with supply chain and logistics systems
2. Discuss the approaches for economic, environmental, and social sustainability in logistics and supply chain
3. Analyse and discuss measures to mitigate negative impacts from logistics and supply chain
4. Identify KPIs for measuring sustainability aspects in logistics and supply chain
5. Develop sustainable solutions for logistics systems
6. Discuss economics implications in logistics and transport
7. Comprehend the impacts of transport economics and policy on other industries and markets
8. Discuss directions for transport policy, infrastructure development and related implications

Course Content:
Environmental, economic and social impacts of logistics, Techniques adopted for assessing the environmental impacts, Sustainability related KPIs, Pollution estimation methods, Negative externalities, Strategies for environmental sustainability, Equity issues in transport, Non-motorised transport, Park and Ride, Ride Sharing, Mobility management, Health, safety and security issues in transport and logistics, Social exclusion, Green supply chain management, Supply chain innovation, Sustainable development goals, Closed-loop supply chain, Reverse logistics, Energy sources, Energy consumption, Emission calculation, Resource allocation, Rules and regulations for environmental sustainability, Transport economy: Utility theory, Opportunity cost, Elasticity of transport demand, Transport supply, Transport pricing and equity, Tax and subsidy, Economic and diseconomy of scale, Generalized costs, Analysis on transport policy, Logistics and transport market structures, Alliances, Transport infrastructure developments, Cost-benefit analysis, Marginal cost and Fixed cost pricing, Discriminatory pricing, Methods of financing transport infrastructure and investments, Investments risks, Ownership issues, Emission tax, Emission trading, Congestion pricing, Benefits/cost analysis of transport projects

Teaching /Learning Methods:
Lectures, interactive classroom sessions and case analysis

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment 40%	Final Assessment 60%		
Details: Assignment%, Project 0 - 20 %, 0 – 20%	Theory (%) 10%- 90%	Practical (%) 0%	Other (%) (Attendance) 0%- 10%

Recommended Reading:

1. Meng, L. and Bock, J.D. (2016) Sustainable Logistics and Supply Chains, Innovations and Integral Approaches, Springer Link.
2. Grant, D.B., Trautrims, A. and Wong, C.Y (2015)Sustainable Logistics and Supply Chain Management (Revised Edition), Kogan Page Publishers.
3. Material provided in the CAL

Year of Study	4		
Semester	2		
Course Code	MGTE 42213		
Course Name	Industrial and Systems Engineering		
Pre-requisites	MGTE 21233 - Operations Management		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Teacher directed learning	Practical	Guided Self-study
	45		105

Course Aim/Intended Learning Outcomes:
On the completion of this course, the student should be able to:

1. Demonstrate understanding of the scope of systems engineering for dealing with work-processes, optimization methods, and risk management tools in large and complex projects.
2. Demonstrate knowledge of value engineering in product and process design.
3. Apply methods for productivity improvement in industrial systems.
4. Analyse the performance of man-machine systems and manage health and safety at work.
5. Identify the opportunities for energy conservation in industry.
6. Analyse the operational and maintenance performance of industrial systems.

Course Content:
Scope of Industrial and Systems Engineering, Engineering of integrated systems, Discrete-time systems, design of work stations and lines, batch processing systems, system performance measures and cost tradeoffs,
System stages in the life-cycle, risk management
Value analysis and value engineering in product design
Method study, Technology for methods improvement in industrial systems, work measurement techniques, Analysis of learning effects, Incentive piece-rate schemes,
Principles of ergonomics, human factors in product and industrial system design,
Industrial health and safety, Accidents and prevention
Improving energy economy in industrial systems, renewable energy applications
Reliability and Plant maintenance

Teaching /Learning Methods:
Lectures, interactive classroom sessions and case analysis

Assessment Criteria:
End-of- Semester examination and case study presentations

Continuous Assessment 25 - 50%	Final Assessment 50 - 75%		
Details: Assignments%, Case analysis% 0 - 50 %, 0 – 50 %	Theory (%) 100%	Practical (%) 0%	Other (%)(specify)

Recommended Reading:

1. Gavriel S. (2001),Handbook of Industrial Engineering, Institute of Industrial Engineering, John Wiley & Sons.
2. Wallace J. Hopp Supply Chain science, McGraw-Hill
3. Adedeji B. Badiru, Handbook of Industrial and Systems Engineering, Taylor and Francis
4. Philips, (2000),Human Factors Engineering, John Wiley & Sons.
5. Campbell, J D, (2001), Maintenance excellence, (Optimising Equipment Life-cycle decision), Marcel Dekker.
6. Material provided in CAL.

Year of Study	4		
Semester	2		
Course Code	MGTE 42223		
Course Name	Investment Management		
Pre-requisites	MGTE 41233 - Corporate Finance		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Independent Study / Practical	Guided Self-study
	45	70	35

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Compare the different investment products available in the market and the risk return profile
2. Use asset allocation criteria to determine investment strategies to meet alternative investment objectives
3. Use valuation techniques to identify mispriced assets
4. Explain the fundamental drivers of diversification
5. Manage the risk and return characteristics of a portfolio
6. Analyze the performance of the portfolio
7. Use forward / futures contracts as a risk mitigation strategy

Course Content:
Introduction to investment and investment analysis, Capital markets and instruments, Securities trading, Risk and risk aversion, Capital allocation between risky and risk free asset, Optimal risky portfolios, Market efficiency, Investment performance analysis, Options, futures and forward contracts;

Teaching /Learning Methods:
Lectures, classroom discussions, group study and guided self-study assignments

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment	Final Assessment		
20 - 60%	40 - 80%		
Details: Quizzes %, mid-term %, Assignments%	Theory (%)	Practical (%)	Other (%)(specify)
0 - 40%, 0 – 40% 0 – 40%	100%	0%	

Recommended Reading:

1. Bodie, Z, Kane, A, Marcus, A J, (2022), Essentials of Investments, 12th Edition, McGraw-Hill/Irwin.
2. Ross S A, Westerfield R W, Jordan B D, (2021), Fundamentals of Corporate Finance, 13th Edition, McGraw-Hill Publishing Company Limited.
3. Hull, J C, (2018), Options Futures and Other Derivative Securities, 10th Edition, Prentice Hall.
4. Daily Financial Times, seen at www.ft.lk
5. Material provided in CAL.

Year of Study	4		
Semester	2		
Course Code	MGTE 42243		
Course Name	Advanced Planning and Scheduling		
Pre-requisites	MGTE 12263 - Optimisation Methods in Management Science		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown	Theory	Tutorial/ Practical	Guided Self-study
(Notional Hrs.)	45	30	75

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:
1. Examine the use of formal methods for operations planning and control.
2. Apply formal and heuristic algorithms for operations planning and control.
3. Use programming for data science applications in planning and scheduling.

Course Content:
Static forecasting models, Winters model, Adaptive forecasting, Box-Jenkin models, Formal optimisation methods for capacity planning and scheduling, Applications of Dynamic programming in production planning, finite scheduling methods, Job scheduling algorithms and sequencing rules, Due date methods, matrix method for project activity scheduling, Heuristics algorithms for lot sizing, dynamic line balancing algorithms, line-of-balance, manpower scheduling, Queueing networks, Decomposition method, Artificial intelligence concepts and applications: A general overview, predictive modelling with machine learning, Metaheuristic optimisation, agent-based modelling and simulation.

Teaching /Learning Methods:
Lectures, interactive classroom sessions and case analysis

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment	Final Assessment	
0 – 40%	0 - 60%	
Details: Assignment%, Project	Theory (%)	Practical (%)
0 - 20 %, 0 – 20%	10%- 90%	0%

Recommended Reading:

1. Chase, R B, Aquilano, N J and Jacobs, F R, (2008), Production and Operations Management – Manufacturing and Services, McGraw-Hill.
2. Nahmias, S, (2010), Production and Operations Analysis, McGraw-Hill International.
3. Anderson, D R, Sweeney, D J and Williams, T A (2010), West Publishing.
4. North, M. J. and Macal, C. M., (2007), Managing Business Complexity: Discovering Strategic Solutions with Agent-Based Modelling and simulation, Oxford University Press.
5. Material provided in CAL.

Year of Study	4		
Semester	2		
Course Code	MGTE 42313		
Course Name	Advanced Supply Chain and Logistics Applications		
Prerequisites	All compulsory modules for OSCM		
Co-requisites	None		
Credit Value	3		
Compulsory/Optional	Compulsory		
Hourly Breakdown (Notional Hrs.)	Theory	Practical/Tutorial	Guided Self-study
	45	15	90

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Plan a logistics system with its components
2. Discuss integration of logistics functions
3. Evaluate factors, different methods and analysis techniques used for logistics system
4. Comprehend advanced concepts in Supply Chain Management
5. Formulate and solve real-life supply chain and logistics related models
6. Develop competency in logistics network planning and distribution networks
7. Rationalize different techniques for managing physical distribution
8. Develop and apply computer-based techniques of distribution planning and control
9. Understand emerging trends, national and international laws and practices in logistics and supply chain

Course Content:
Logistics system, Logistics activity profiles, Short, medium, and long term planning for logistics infrastructure, Demand planning, Scheduling, Forecasting, Facility design and operations, Demand and order management, Advanced Inventory models, Logistics systems integration, Reverse logistics systems, Outsourcing levels - Third Party Logistics (3PL) Provider / Lead logistics Provider (LLP) (Linking to Logistics Ecosystems, 2PL, 4PL, 5PL, Drop shipping, Last Mile Logistics), Supply chain restructuring, Decision levels in a supply chain, Tripple A supply chain, Freight forwarding, Decision making for logistics systems, Derived demand in transport, Transport supply, Infrastructure planning, Design and layouts of infrastructure, Investments in transport infrastructure, Maritime, Aviation, Land transport, Transport Demand and Supply, Formulation of mathematical models for solving issues in logistics systems, Aggregate and disaggregate planning, Facility location problems, Logistics network planning, Green Supply Chain, Closed-loop supply chain, Introduction to Physical Distribution: Interfaces & environment; functional areas; Systems design & measurement; Decision making; Distribution planning and scheduling, Organization and administrative costing; Components of physical distribution systems, Distribution Requirements Planning (DRP)systems, analysis and benefits of DRP; Distribution networks, Emerging trends in logistics, National and international law in logistics, legislation, regulations, safety requirements, and professional standards

Teaching /Learning Methods:
Lectures and case discussions

Assessment Criteria:
End of course unit examination and continuous assessments.

Continuous Assessment	Final Assessment		
25 - 50%	50 - 75%		
Details: quizzes %, mid-term %, Assignments% 0 - 50 %, 0 – 50 % 0 – 50 %	Theory (%) 100%	Practical (%) 0%	Other (0-10%) (attendance)

Recommended Reading:

1. Ghiani, G., Laporte, G. and Musmanno, R. (2004) Introduction to Logistics Systems Planning and Control
2. Langevin, A. and Riopel, D. (2005) Logistics Systems: Design and Optimization
3. Farahani, R.Z., Rezapour, S. and Kardar, L. (2011) Logistics Operations and Management, Concepts and Models, Elsevier Inc.
4. Rahul, V A, (2005), Supply Chain Management and Cases, PHI Learning, New Delhi.
5. Chopra, S and Peter, M (2013) Supply Chain Management, Strategy, Planning, And Operation. Pearson
6. Joel, D W, Keong, L and Keah Choon, T, (2012), Principle of Supply Chain Management – A Balanced Approach, Cengage Learning.
7. Ronald, H B, Samir, K S, (2007), Business Logistics/ Supply Chain Management, Pearson Education.
8. Paul, R M and Donald, F W, (2008), Contemporary Logistics, PH of India.
9. Material provided in CAL.

Year of Study	4		
Semester	2		
Course Code	MGTE 42323		
Course Name	Strategic Quality Management and Lean Six Sigma		
Pre-requisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	45	15	90

Course Aim/Intended Learning Outcomes:
On completion of this course, the student should be able to:
1. Introduce quality management concepts and techniques.
2. Apply Six Sigma and lean concepts applicable in real world settings to improve effectiveness, efficiency and productivity.
3. Acquire Six Sigma Professional Certification.

Course Content:
Introduction to Quality Concepts, Six Sigma method, DMAIC Approach, Reliability and Survival Analysis: Gage Repeatability& Reproducibility (Gage R & R), Design and Analysis of Experiments: The Design & Analysis of Factorial Experiments for 2 Factors, The Design and Analysis of Factorial Experiments with 3 or More Factors, Lean Concepts in Production and Operations Management, Value Stream Management (VSM), Quality Management Philosophies, Tools and Techniques in Quality Management, 7 Basic Quality Management tools, 7 New Quality Management tools, Statistical Process Control: Basic Tools and Principles of Process Control, Control charts, Multivariate Statistical Process Control, Process Capability, Total Quality Management (TQM), Quality Management Principles for Performance Improvement, ISO Standards and Quality Auditing, Quality Inspections, Quality Risks and Mitigation, Cost of Quality and Issues in Quality Management

Teaching /Learning Methods:
Lectures, Assignments, Case studies and self-study

Assessment Criteria:
End-of-semester examination, Mid-semester exam, Case analysis and Presentations

Continuous Assessment	Final Assessment
40%	60%
Details: In class Assessments 10%, Group Assignments 30%,	

Recommended Reading:

1. Pascal, D, (2007), Lean Production Simplified, 2nd Edition, Productivity Press.
2. Womack and Jones, (2005), Lean Solutions: How Companies and Customers Can Create Value and Wealth Together, 1st Edition, Free Press.
3. Howard, S G and David, M L, Six Sigma for Green Belts and Champions: Foundations, DMAIC, Tools, Cases, and Certification, 1st Edition, PH Professional Business.
4. Donna C S, (2006), Six Sigma: Basic Tools and Techniques, 1st Edition, Prentice Hall.
5. Jeffrey K. Liker (2004) The Toyota Way: 14 Management Principles from the World's Greatest Manufacturer
6. Material provided in CAL.

Year of Study	4		
Semester	2		
Course Code	MGTE 42333		
Course Name	Business and Information Technology Law		
Pre-requisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	45	15	90

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Identify the necessity of legal frameworks for business organizations.
2. Demonstrate an understanding of the basic concepts of business law.
3. Explain the new legal challenges faced by business firms in the Information Technology era.
4. Explain the key acts and laws available in Sri Lanka that are related to business and information technology.
5. Discuss the barriers available in the Sri Lankan legal system to deal with the development of ICT.

Course Content:
The need for laws to govern business and ICT; Basic concepts of business law – contracts, torts, business structures; Legal challenges of ICT in business; The existing ICT related laws in Sri Lanka – Laws related to contracts (EULAs, e-Signatures, contract formation), Electronic commerce, Electronic data, Cybercrimes; Intellectual Property related laws in Sri Lanka and their adequacy to meet the new challenges in the Digital Age; Status of laws relating to Tort with development of ICT (Ex. Social media); Challenges arising from emerging technologies – Dealing with artificial intelligence and robotics, Health hazards, Spying

Teaching /Learning Methods:
Lectures, case analysis and guided self-study assignments

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment 25 – 40%	Final Assessment 60 – 75%		
Details: quizzes %, mid-term %, Assignments% 0 - 40 %, 0 – 40 % 0 – 40 %	Theory (%) 100%	Practical (%) 0%	Other (%)(specify)

Recommended Reading:

1. Andrew Murray, (2019), Information Technology Law, 4th Edition.
2. I. Lloyd, (2017), Information Technology Law, 8E, Oxford University Press.
3. Sri Lanka Business Law Handbook, Vol. 1, Strategic Information and Basic Laws, 6E, (2019), International Business Publications.
4. Materials provided in CAL.

Year of Study	4		
Semester	1 AND 2		
Course Code	MGTE 43216 / INTE 43216		
Course Name	Research Project		
Pre-requisites	MGTE 31383 - Research Methods		
Co-requisites	None		
Credit Value	6		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
	0	0	600

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. explain the scientific process of carrying out a research
2. identify and formulate a suitable research problem in the fields of Management and/or Information Technology
3. design and carry out a survey of literature, conduct experiments, formulate solutions/models, develop porotypes, collect and analyse data
4. critically analyse research findings
5. summarize research findings and write research publications

Course Content:
Each student is assigned a research project by the department in the fields of Management and/or Information Technology, directly relevant to a typical industry issue and the students are expected to research and apply the knowledge gained throughout the degree programme in solving assigned research problem

Teaching /Learning Methods:
Progress review meetings, a series of seminars, presentations and discussions.

Assessment Criteria:
Research proposal evaluation, mid-year progress review assessment and end year dissertation evaluation.

Continuous Assessment 60%	Final Assessment 40%		
Details: research proposal, mid-year progress review, publications 20%, 25%, 15%	Theory (%) 0%	Practical (%) 0%	Other (%) (dissertation evaluation) 100%

Recommended Reading:

1. Keerthi Wijayasiriwardhane, Suren Peter and Annista Wijayanayake, (2017), Study Guide for Research Project, Department of Industrial Management, Faculty of Science, University of Kelaniya.
2. Colin Robson (2016), How to do a research project - A guide for undergraduate students, 2nd edition, Wiley.
3. Material provided in CAL.

Year of Study	4		
Semester	1 OR 2		
Course Code	MGTE 44273		
Course Name	Innovation and Entrepreneurship		
Pre-requisites	None		
Co-requisites	None		
Credit Value	3		
Hourly Breakdown (Notional Hrs.)	Theory	Independent Study / Practical	Guided Self-study
	45	35	70

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Discuss the attitudes, values, characteristics, behaviour, and processes associated with possessing an entrepreneurial mindset and engaging in successful appropriate entrepreneurial behaviour.
2. Differentiate between innovation and entrepreneurship from both a theoretical and practical perspective, and the role of the entrepreneur in the new enterprise creation process.
3. Describe the ways in which entrepreneurs perceive opportunity, manage risk, organise resources and create a unique value
4. Generate an innovative business model for a lean start-up, utilizing a structured set of tools
5. Evaluate potential risks for start-ups and business transformations
6. Create a comprehensive business plan and a lean start-up pitch-deck incorporating the novel business model, financials and risk mitigation plans.

Course Content:
Entrepreneurship: Entrepreneurial process, Entrepreneurial characteristics, Sustainability, Managing resources, finances and risks.
Innovation: Types of innovation, Design thinking, Types of new products; New products & Product Life Cycle; Tools for innovation management (design for manufacturing & assembly (DFX), quality function deployment) Concept development; Product positioning; Assessing market potential, Business model
Creating an Innovative organization: Innovative enterprise, Innovation diffusion & knowledge management, dynamic capabilities, leadership, organizational innovation
Market your idea: Business plan & Pitch deck, Lean start-up, Product and market testing; Product launch

Teaching /Learning Methods:
Lectures, supervised practical sessions and guided self-study assignments

Assessment Criteria:
End of course unit examination and continuous assessments

Continuous Assessment 25 - 50%	Final Assessment 50 - 75%		
Details: quizzes %, mid-term %, Assignments% 0 - 50 %, 0 – 50 % 0 – 50 %	Theory (%) 100%	Practical (%) Develop a business proposal	Other (%)

Recommended Reading:

1. Paul, T, (2010), Innovation Management and New Product Development, 4th Edition, Pearson.
2. Ettlie, J E, (2010), Managing Innovation: New technology, new products and new services on a global economy, 2nd Edition, Butterworth-Heinemann.
3. Tidd, J and Bessant J, (2009), Managing Innovation: Integrating Technological, Market and Organizational Change, 4th Edition, John Wiley & Sons.
4. Ries E, (2011), The Lean Startup, 1st Edition, Penguin.
5. Christensen, C. M., (1997), The innovator's dilemma: When new technologies cause great firms to fail. Boston, Mass: Harvard Business School Press.
6. Reis, E, Managing Innovation: The Lean start up.
7. Theil P., (2014), Zero to One, Crown Business.
8. Material provided in CAL.

c) Mathematics

Year of study 1:

Year of Study	1		
Semester	1		
Course Code:	PMAT 11212		
Course Name:	Discreet Mathematics for Computing - 1		
Credit Value:	2		
Pre-Requisites	None		
Co-Requisites	None		
Hourly Breakdown	Theory	Practical/ Tutorial	Self-Guided Study
	30	15	55

Course Aim/Intended Learning Outcomes:
At the completion of this course student will be able to:
1. Demonstrate knowledge in logic, sets, relations, and functions
2. Apply Boolean algebra to simplify combinatorial circuits

Course Content:
Propositional Logic - Propositions, Truth values, Logical connectives, Truth table, Tautology and Contradiction, Logical equivalence, Algebra of propositions, Validity of an argument; Predicate Logic - Quantifiers, Nested quantifiers, Negation of quantified statements, Validity of an argument with quantifiers; Methods of Proof - Informal idea of a theorem and a proof, Converse, inverse and the contrapositive of a statement, Direct proof, Proof by contradiction, contrapositive, exhaustion and cases, Disproving by counter-examples, Principle of mathematical induction (weak and strong form); Sets - Set notations, Sets of numbers, Subsets of the real numbers and interval notation, Operations on sets, Algebra of sets, Set identities, Power set, Cartesian product of sets; Relations - Equivalence relations and equivalence classes, Properties of equivalence classes, Partitioning of sets; Functions - Function notations, Image and pre-image, One-to-one and onto functions, Composition of functions, Inverse Function, Image and inverse image of subsets under functions; Boolean algebra - Axioms of Boolean algebra and its properties, Correspondence between Boolean algebra and combinatorial logic circuits, Simplifications of combinatorial logic circuits using Boolean algebra.

Teaching /Learning Methods:
A combination of lectures and tutorial discussions

Assessment Strategy:
Based on the tutorials, tests, and end of course examination

Continuous Assessment 20 %	Final Assessment 80 %		
Details: Tests 50%, Assignments 0 %, Tutorials 50%	Theory (%) 100	Practical (%) 0%	Other (%) (specify) N/A

References/Reading Materials:

4. Johnsonbaugh, R., (1990), Discrete Mathematics, MacMillan.
5. Lipschutz, S., (1976), Discrete Mathematics, McGraw-Hill.
6. Materials provided in the CAL.

Year of Study	1		
Semester	2		
Course Code:	PMAT 12212		
Course Name:	Discreet Mathematics for Computing - 2		
Pre-Requisites	PMAT 11212 Discreet Mathematics for Computing - 1		
Co-Requisites	None		
Credit Value:	2		
Hourly Breakdown	Theory	Practical/ Tutorial	Self-Guided Study
	30	15	55

Course Aim/Intended Learning Outcomes:
At the completion of this course student will be able to:
1. Demonstrate knowledge in matrices and counting techniques
2. Apply matrices to analyze systems of linear equations
3. Use counting techniques in modeling real world problems
4. Demonstrate knowledge in basic graph theory

Course Content:
Matrices - Matrix notations, Algebra of matrices, Inverse of a matrix, Elementary row operations and calculation of inverse matrix, Determinant of a matrix, Minor and cofactor, Properties of determinants, Adjoint of a matrix and calculation of inverse matrix; Systems of Linear Equations - Matrix form of a system of equations and conditions for unique solutions, augmented matrix and echelon form, Conditions for existence of a unique solution, infinitely many solutions and no-solution; Counting Techniques - Factorial notation, Sum rule and product rule, Permutations and combinations, Binomial expansion; Graphs - Graph notation and basic definitions, Complete and bipartite graphs, Euler and Hamiltonian circuits, Directed and undirected graphs, Trees and their basic properties

Teaching /Learning Methods:
A combination of lectures and tutorial discussions

Assessment Strategy:
Based on the tutorials, tests, and end of course examination

Continuous Assessment	Final Assessment		
20 %	80 %		
Details:	Theory (%)	Practical (%)	Other (%) (specify)
Tests 50%, Assignment 0 %, Tutorials 50%	100	0	N/A

References/Reading Materials:

1. Susanna, S. E., (2010), Discrete Mathematics with Applications, Cengage Learning.
2. Kenneth, H. R., (2007), Discrete Mathematics and Its Applications, McGraw-Hill.
3. Materials provided in the CAL.

Year of study 3:

Year of Study	3		
Semester	1		
Course Code:	PMAT 31212		
Course Name:	Mathematics for Computing - 3		
Pre-Requisites	PMAT 11212 Discreet Mathematics for Computing - 1, PMAT 12212 Discreet Mathematics for Computing 2		
Co-Requisites	None		
Credit Value:	2		
Hourly Breakdown	Theory	Practical/ Tutorial	Self-Guided Study
	30	15	55

Course Aim/Intended Learning Outcomes:
At the completion of this course student will be able to:

1. Demonstrate knowledge in basic mathematical concepts of calculus.
2. Use basic mathematical concepts of calculus for further studies.
3. Categorize ordinary differential equations.
4. Solve linear ordinary differential equations using appropriate methods.
5. Derive the ordinary differential equations for certain applications.

Course Content:
Limits and Derivatives: Limit of a Function, Calculating Limits Using the Limit Laws, Continuity, Limits at Infinity-Horizontal Asymptotes, Derivatives and Rates of Change, Derivative as a Function, Differential rules, Applications of Differentiation
Integrals: Definite Integral, Fundamental Theorem of Calculus, Substitution Rule, Applications of Integration: Areas between Curves, Volumes, Volumes by Cylindrical Shells, Arc Length, Area of a Surface of Revolution, Techniques of Integration
First-Order Ordinary Differential Equations (ODEs): Separable ODEs. Exact ODEs. Integrating Factors, Linear ODEs, Higher Order Linear ODEs: Homogeneous Linear ODEs, Homogeneous Linear ODEs with Constant Coefficients, Differential Operators, Special types of ODEs: Bernoulli Equations, Euler–Cauchy Equations, Applications of ODEs.

Teaching /Learning Methods:
Lectures, interactive classroom sessions, and case discussions

Assessment Strategy:
End of course unit examination, group assignment, mid-term examination, class attendance

Continuous Assessment 20 %	Final Assessment 80 %		
Details: Tests 50%, Assignments 50 %, Tutorials 0%	Theory (%) 100	Practical (%) 0%	Other (%) (specify) N/A

References/Reading Materials:

1. Kreyszig, E., (2011), 10th edition, Advanced Engineering Mathematics, Wiley.
2. Ross, K.S., (2015), 2nd edition, Elementary Analysis: The Theory of Calculus, Springer.
3. Stewart, J., (2015), 8th edition, Calculus Early Transcendentals, Thomson Learning, Inc.
4. Materials provided in the CAL.

d) Communication Competencies

Year of Study	1		
Semester	1		
Course Code	DELT 11232		
Course Name	English for Professionals		
Pre-Requisites	None		
Co-Requisites	None		
Credit Value	2		
Hourly Breakdown (Notional Hrs.)	Theory	Practical/ Tutorial	Self-Guided Study
	30	0	70

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Apply professional etiquette practices in business contexts
2. Use the right linguistic strategies to listen and respond critically
3. Face interviews with confidence
4. Identify the rhetoric and style of CV, resume and cover letters
5. Write persuasively and professionally

Course Content:
Introduction to business language, Professional etiquette(Giving requests, advice, recommendations/offers, instructions, orders etc), CV, resume and cover letter writing, Interview skills, Active listening and critical thinking.

Teaching /Learning Methods:
Interactive classroom sessions, Model Interviews and meetings, workshops and peer review sessions

Assessment Criteria:
End of course unit examination and continuous assessments

2 Continuous Assessments	Final Assessment
40%	60%

Recommended Reading:

1. Folse, K S, Mahnke, M K, Solomon, E V and Williams, L (2003). Blueprints 1: Composition Skills for Academic Writing. New York: Houghton Mifflin Company.
2. Lefevre, C A and Lefevre, H E (1984). Reading Power and Study Skills for College Work. New York: Harcourt Brace Jovanovich, Inc.
3. Goodale, M (1998). Professional Presentations. Cambridge: Cambridge University Press.
4. Materials provided in the CAL.

e) General Competencies

Year of Study	1		
Semester	1		
Course Code	GNCT 11212a		
Course Name	Personal Progress Development I		
Pre-Requisites	None		
Co-Requisites	None		
Credit Value	2		
	(Credits will not be calculated for the GPA)		
Hourly Breakdown (Notional Hrs.)	Theory	Practical/ Tutorial	Self-Guided Study
	15	15	70

Course Aim/Intended Learning Outcomes:
After completing this module, the students should be able to,

1. Demonstrate knowledge and competence in key skills required in personal and professional development and have developed a high degree of self–awareness
2. Evaluate their own professional and personal development experiences.
3. Plan and manage their professional development through effective self-directed learning and reflection on their own practice and experience.
4. Demonstrate advanced interpersonal skills in conducting effective presentations, team working, leadership, professional negotiations and communication both orally and in writing.
5. Develop the capacity of continuous learning and adaptability to perform effectively in different organizational and culture contexts.
6. Demonstrate understanding of aspects of good leadership.
7. Demonstrate understanding of team work in organizations and need to align/manage interests and efforts of diverse groups to meet organizational objectives.
8. Demonstrate improved levels of communication, leadership and team capabilities and professional attributes essential for success in the global business environment.

Course Content:
Introduction on the course unit and Getting to know each other, Group decision making, Individual decision making, Communication and negotiation, Individual goal setting, Mind mapping, Instructions for the poster presentation, Poster presentation, Strategic thinking, Leadership and Team Building, Time management, Paper answering, Report Writing, academic writing I, Trust building, Strategic group decision making, ; Team building; Presentation skills I; Time management; First-aid and Life support; Social and physical environmental sensitivity

Teaching /Learning Methods:
Lectures, in-class discussions and activities, real world case studies and self-study

Assessment Criteria:
Continuous assessments

Continuous Assessment 0-100%	Final Assessment 0-100%		
Details: project report %, presentation %, role play %, spontaneous assignment % 0 - 25 %, 0 – 25 %, 0 – 30 %, 0 - 20 %	Theory (%) 0%	Practical (%) 0%	Other (%) (In-Class activities and attendance) 0-90%, 0-10%

Recommended Reading:

1. L T. S. Grover and S. Wenk, Relentless: From Good to Great to Unstoppable, Scribner, 2014
2. R. Sharma, Who Will Cry When You Die? (1999), HarperCollins Publishers.
3. N. M. Tichy and Andrew McGill, The Ethical Challenge: How to Lead with Unyielding Integrity, Jossey-Bass, 1E, (2003).
4. R. Covey, The Seven Habits of Highly Effective Teens, Simon & Schuster, 2014
5. Walter, R (2006), Kick Start Your Success – Four Powerful Steps to Get What You Want out of your Life, Career and Business, John Wiley & Sons.
6. George, K (2006), Coaching into Greatness – Four Steps to Success in Business & Life, John Wiley & Sons.
7. Razeyhi, A, (2006), Hope – How Triumphant Leaders Create the Future, Jossey Bass.
8. Materials provided in the CAL.

Year of Study	2		
Semester	4		
Course Code	GNCT 24212a		
Course Name	Personal Progress Development II		
Pre-Requisites	None		
Co-Requisites	None		
Credit Value	2		
	Credits will not be calculated for the GPA		
Hourly Breakdown (Notional Hrs.)	Theory	Tutorial/ Practical/ In-Class Activities	Guided Self-study
	15	15	70

Course Aim/Intended Learning Outcomes:

After completing this module, the students should be able to,

1. Develop knowledge of and competence in key skills required in personal and professional development
2. Demonstrate a high degree of self–awareness and be able to evaluate their own professional and personal development experiences.
3. Demonstrate their professional development through effective self-directed learning and reflection on their own practice and experience.
4. Demonstrate advanced interpersonal skills in conducting effective presentations, team work, leadership, professional negotiations and communication both orally and in writing.
5. Develop the capacity of continuous learning and adaptability to perform effectively in different organizational and culture contexts.
6. Demonstrate improved levels of professional attributes essential for success in the global business environment.
7. Demonstrate the understanding of taking care of physical and mental health.

Course Content:

Course Introduction and Peer review, CV preparation and facing interviews, Motivation, Mock Interviews, Presentation skills II, Individual Presentation, Group Presentations, Academic writing II, Personality development & Personal branding, Stress Management- Meditation based session, Movie Review, Cultural Communication, Entrepreneurship & Journey of an entrepreneur, Self-Reflection, Balance mind and heart- Taking care of mental health, Emotional Intelligence and Multiple Intelligences, Food & Nutrients, Communication Skills, Leadership , Be Beautiful -Personal Grooming, Career Management , Business Etiquettes & Dining Etiquettes, Event Management- CREST Programme

Teaching /Learning Methods:

Lectures, in-class discussions, real world case studies and self-study

Assessment Criteria:

Continuous assessments

Continuous Assessment 0-100%	Final Assessment 0-100%		
Details: project report %, presentation %, role play %, spontaneous assignment % 0 - 25 %, 0 – 25 %, 0 – 30 %, 0 - 20 %	Theory (%) 0%	Practical (%) 0%	Other (%) (In-Class activities & attendance 0-100%

Recommended Reading:

1. Shepherd, Ifan D. H., (2005), From cattle and coke to Charlie: meeting the challenge of self-marketing and personal branding. Journal of Marketing Management, 21 (5-6). pp. 589-606. ISSN 1472-1376
2. Needle, David. - Business in context: an introduction to business and its environment
3. The Seven Habits of Highly Effective People: Restoring the Character Ethic" by Stephen R. Covey, Simon and Schuster.
4. L T. S. Grover and S. Wenk, (2014), Relentless: From Good to Great to Unstoppable, Scribner.
5. R. Sharma, (1999), Who Will Cry When You Die?, HarperCollins Publishers.
6. N. M. Tichy and Andrew McGill, (2003), The Ethical Challenge: How to Lead with Unyielding Integrity, Jossey-Bass, 1E.
7. R. Covey, (2014), The Seven Habits of Highly Effective Teens, Simon & Schuster.
8. Poverty and Youth Issues in Sri Lanka by Claudia Ibargüen, http://www.cepa.lk/content_images/publications/documents/Poverty-and-Youth-Issues-in-Sri-Lanka-english-20160328165817.pdf
9. Shepherd, Ifan D. H., (2005), From cattle and coke to Charlie: meeting the challenge of self-marketing and personal branding. Journal of Marketing Management, 21 (5-6). pp. 589-606. ISSN 1472-1376
10. Materials provided in the CAL.

Year of Study	3		
Semester	2		
Course Code	GNCT 32216		
Course Name	Internship		
Pre-Requisites	All previous compulsory modules		
Co-Requisites	None		
Credit Value	6		
Hourly Breakdown (Notional Hrs.)	Theory	Practical	Guided Self-study
		600	

Course Aim/Intended Learning Outcomes:
On completion of this industrial training, the student should be able to:

1. Relate concepts learned in the class to organizational and environmental management
2. Integrate and apply the theory, knowledge, skills and values acquired through their first and second years in areas related to their interests and learning needs
3. Identify the issues in making concepts operational, and acquire skills in resolving these issues
4. Perform a range of work skills developed during the training
5. Employ a range of soft skills
6. Utilize the networks developed to advance their career opportunities

Course Content:
Development of learning needs and objectives, within the framework of industrial training and student expectations, Pre-training orientation: work ethics, professional conduct, Placement organization: history, products, organizational goals, organizational structure, personnel profile, operational policies, values, function and place of organization in the community, Exposure and participation in organizational processes, Identification and analysis of suitable industrial problems

Teaching /Learning Methods:
Exposure to world of work, active participation in organizational processes, self-study, analysis of industrial problems, interactive group discussions and presentations

Assessment Criteria:
Fortnightly progress reports, final report, oral presentation and feedback from industrial training supervisor

Presentation 40%	Report 30%	Progress and Submissions 30%

Recommended Reading:

1. Industrial training handbook – B.Sc.MIT, Department of Industrial Management, Faculty of Science, University of Kelaniya.
2. Material provided in the CAL.